

# BFS Traversal

Step 1: Start at A

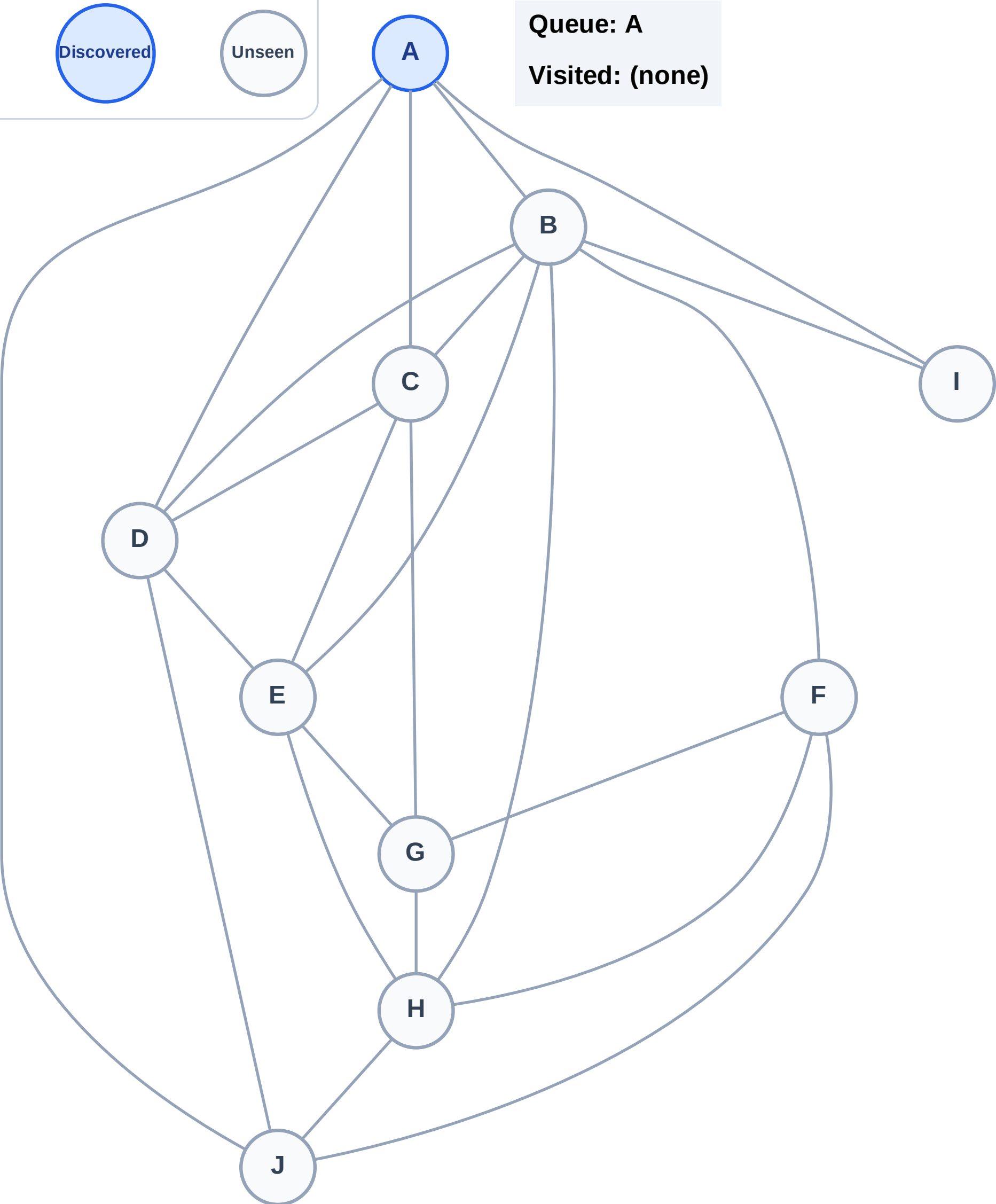
Complete

Processing

Discovered

Unseen

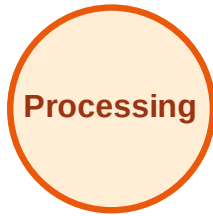
Queue: A  
Visited: (none)



# BFS Traversal

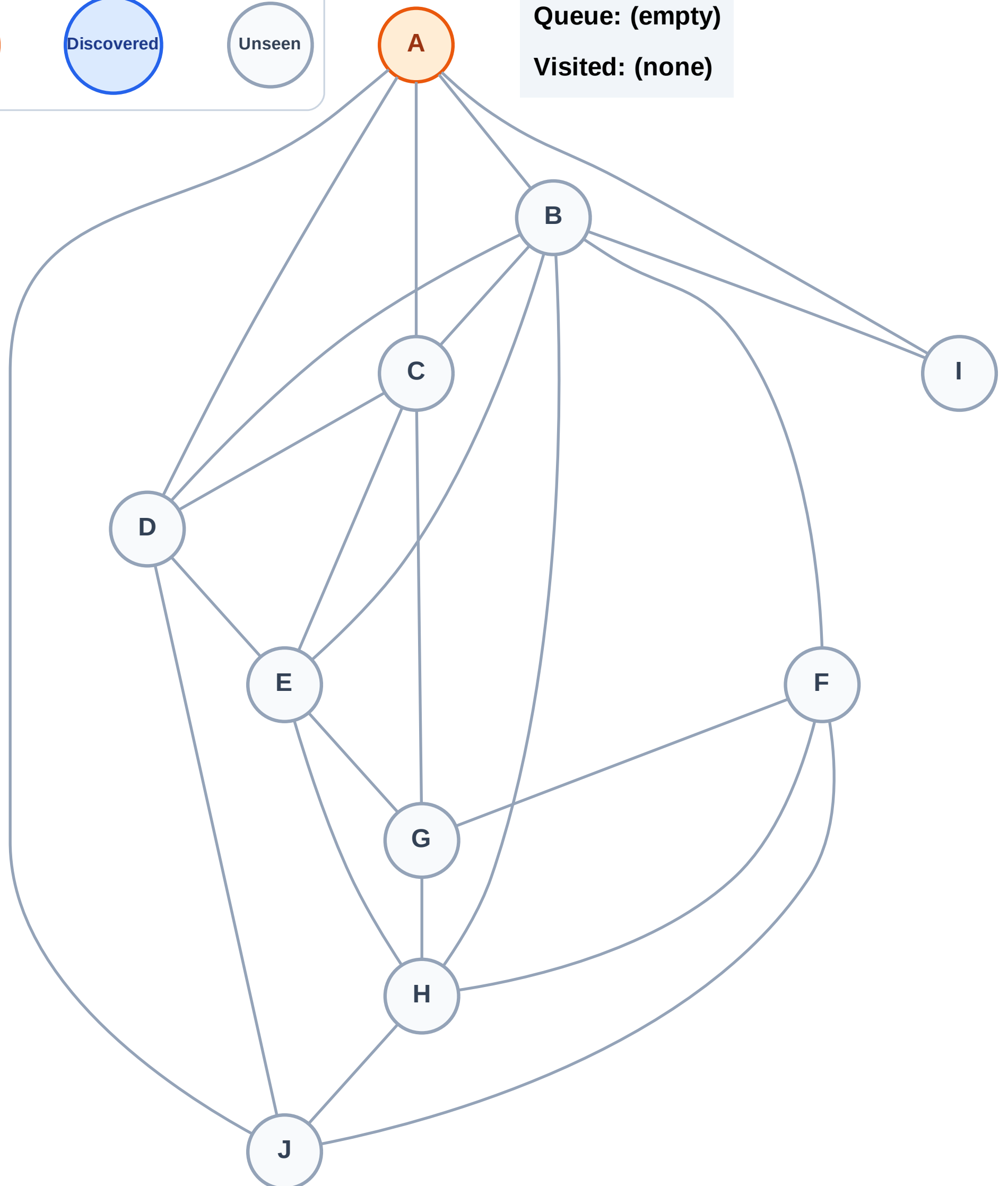
Step 2: Process node A

## Legend



Queue: (empty)

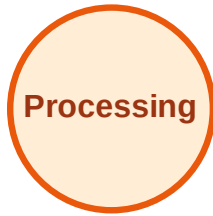
Visited: (none)



# BFS Traversal

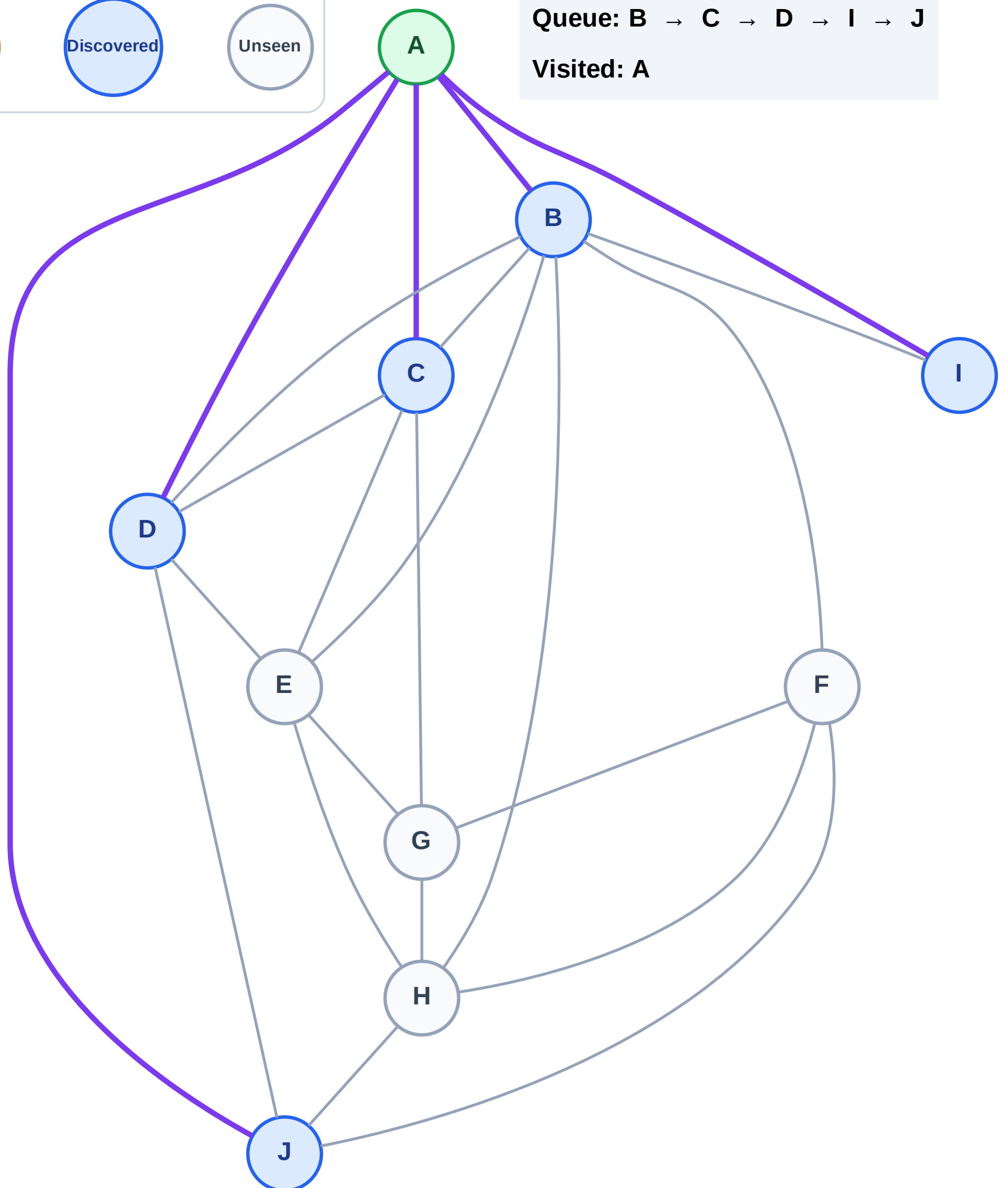
Step 3: Discover B, C, D, I, J from A

## Legend



Queue: B → C → D → I → J

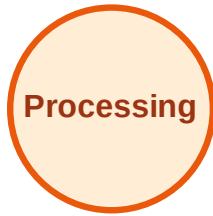
Visited: A



# BFS Traversal

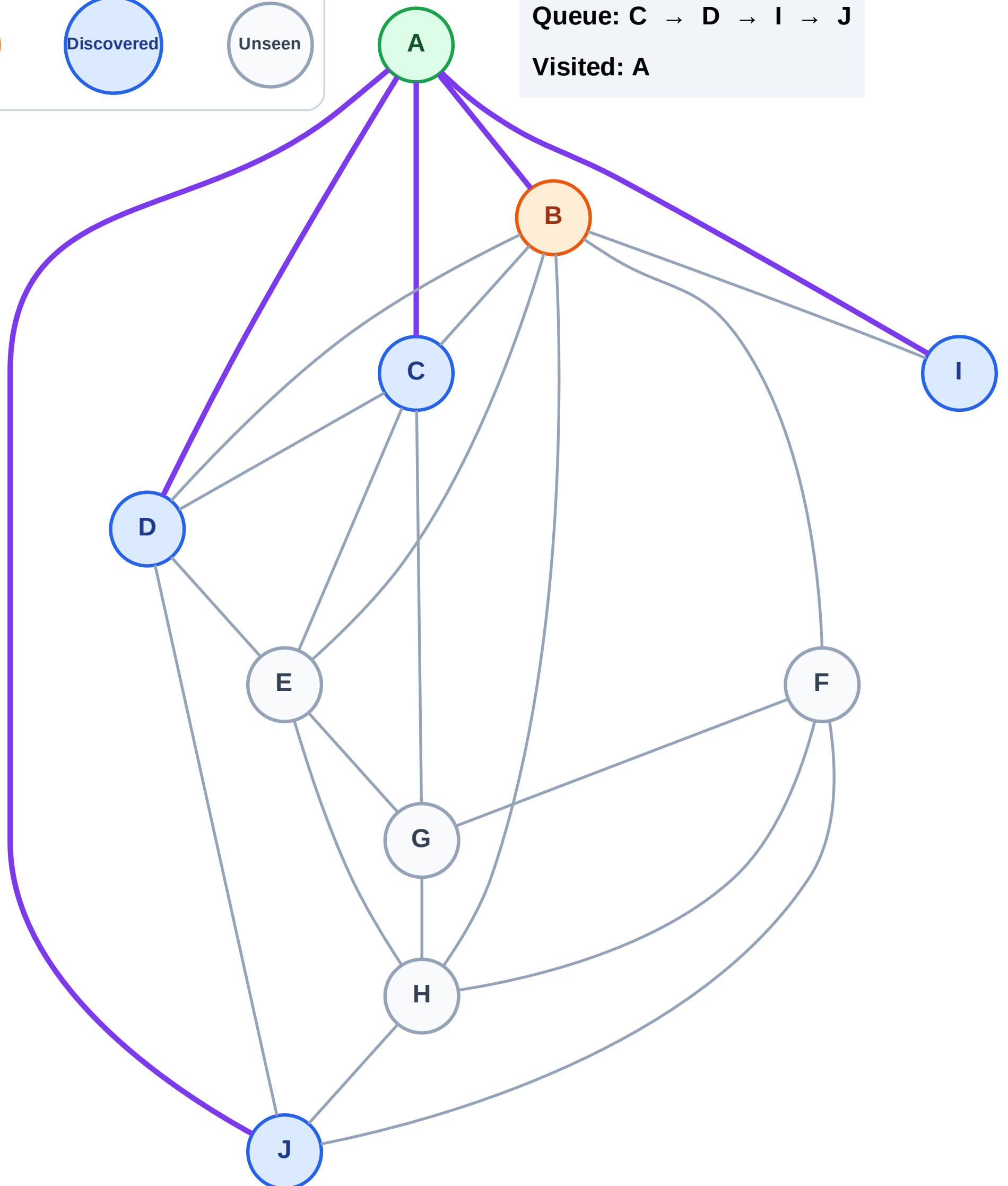
Step 4: Process node B

## Legend



Queue: C → D → I → J

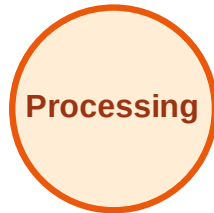
Visited: A



# BFS Traversal

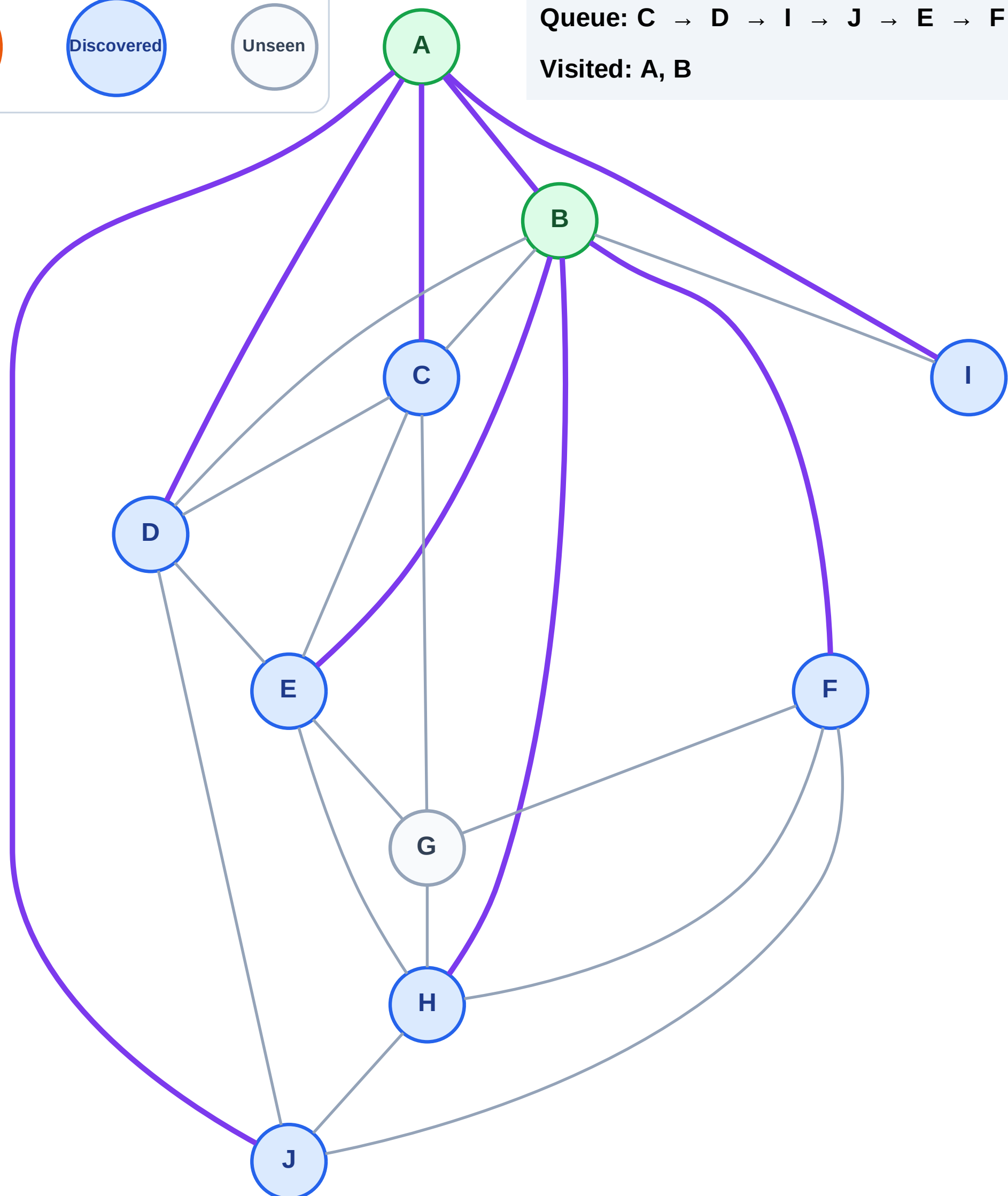
Step 5: Discover E, F, H from B

## Legend



Queue: C → D → I → J → E → F → H

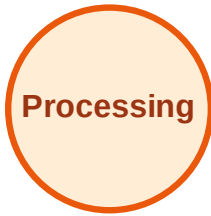
Visited: A, B



# BFS Traversal

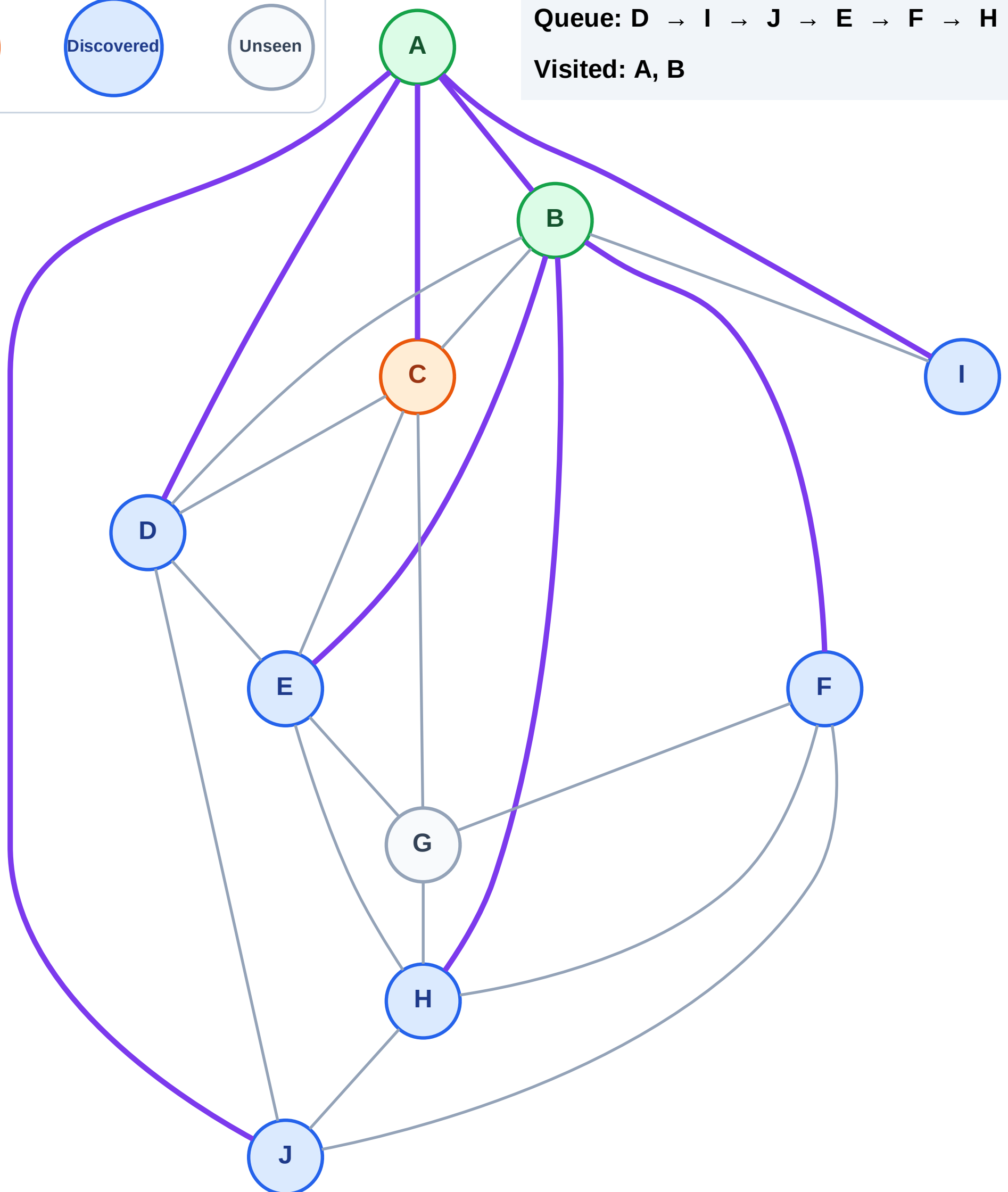
Step 6: Process node C

## Legend



Queue: D → I → J → E → F → H

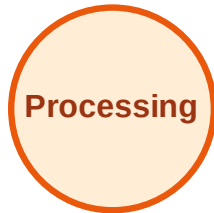
Visited: A, B



# BFS Traversal

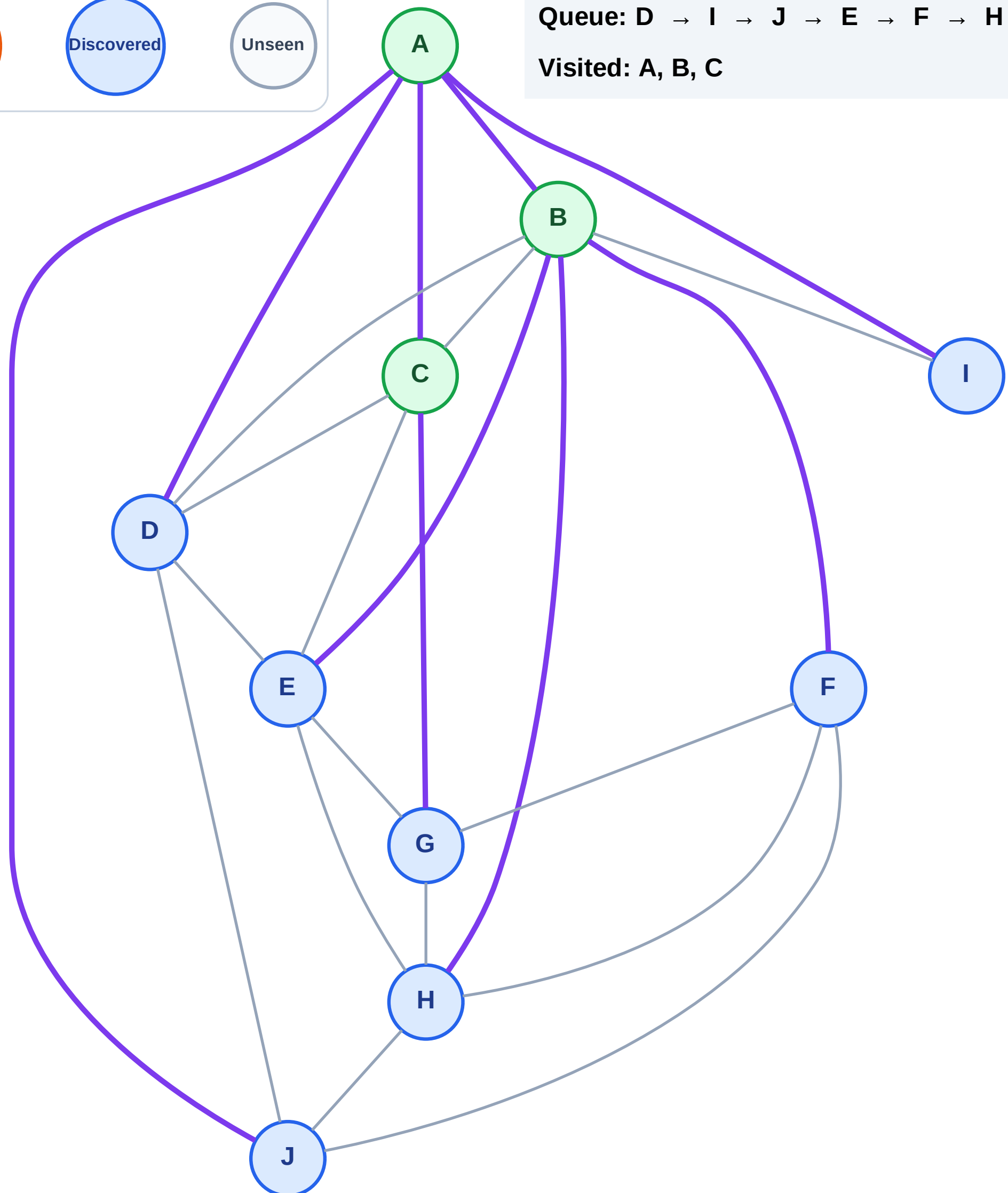
Step 7: Discover G from C

## Legend



Queue: D → I → J → E → F → H → G

Visited: A, B, C



# BFS Traversal

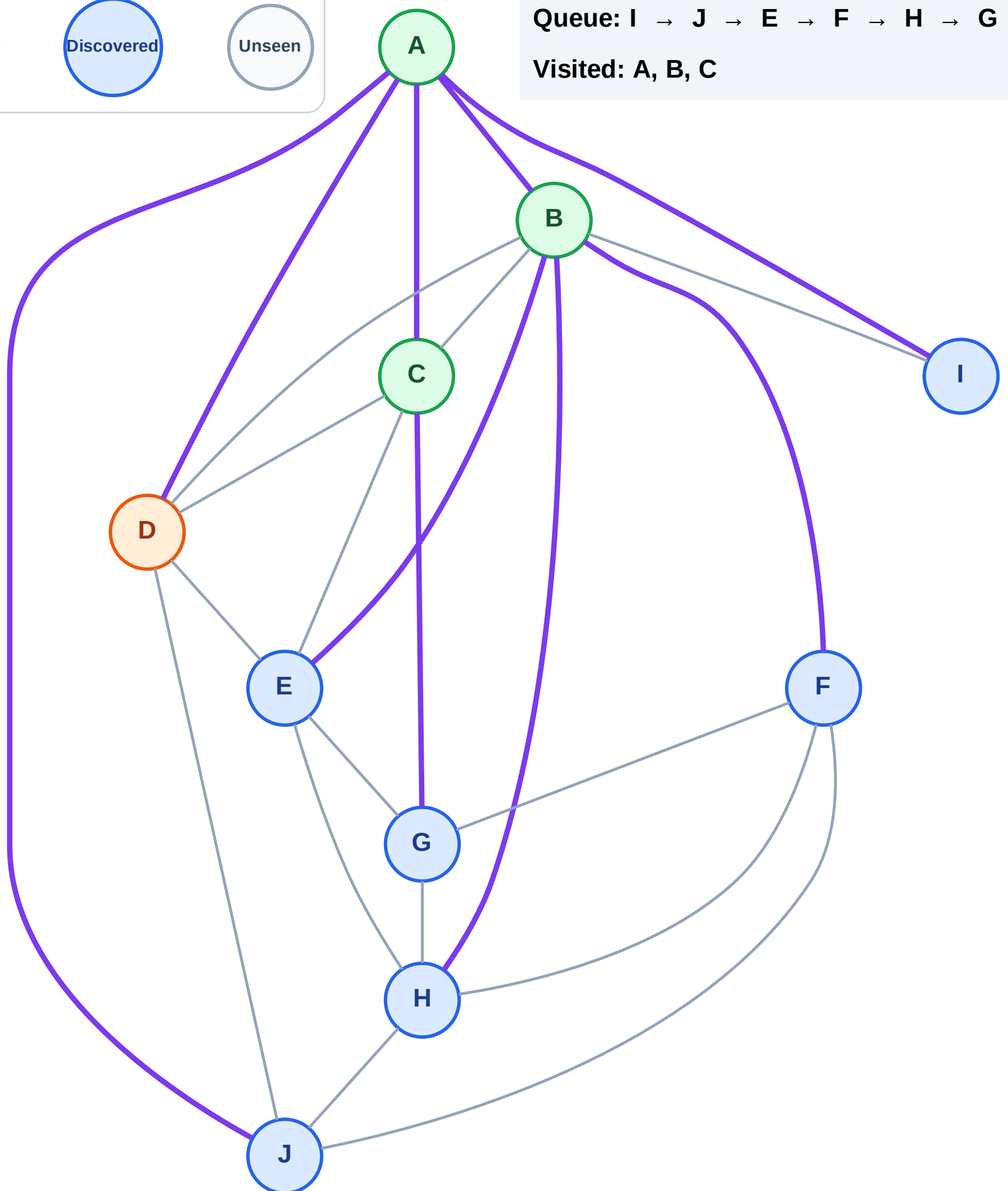
Step 8: Process node D

## Legend



Queue: I → J → E → F → H → G

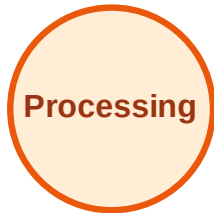
Visited: A, B, C



# BFS Traversal

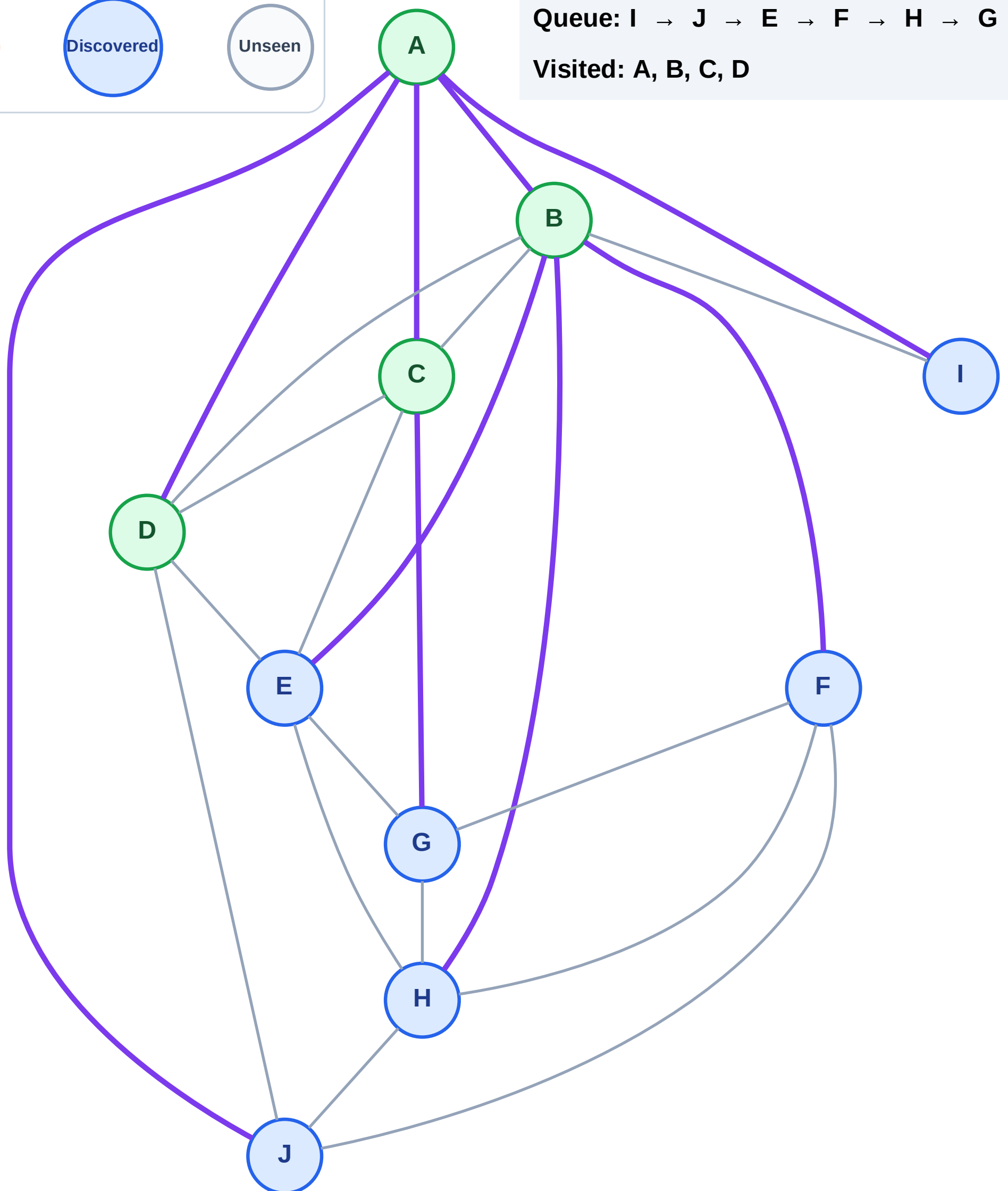
Step 9: No new neighbors from D

## Legend



Queue: I → J → E → F → H → G

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node I

Complete

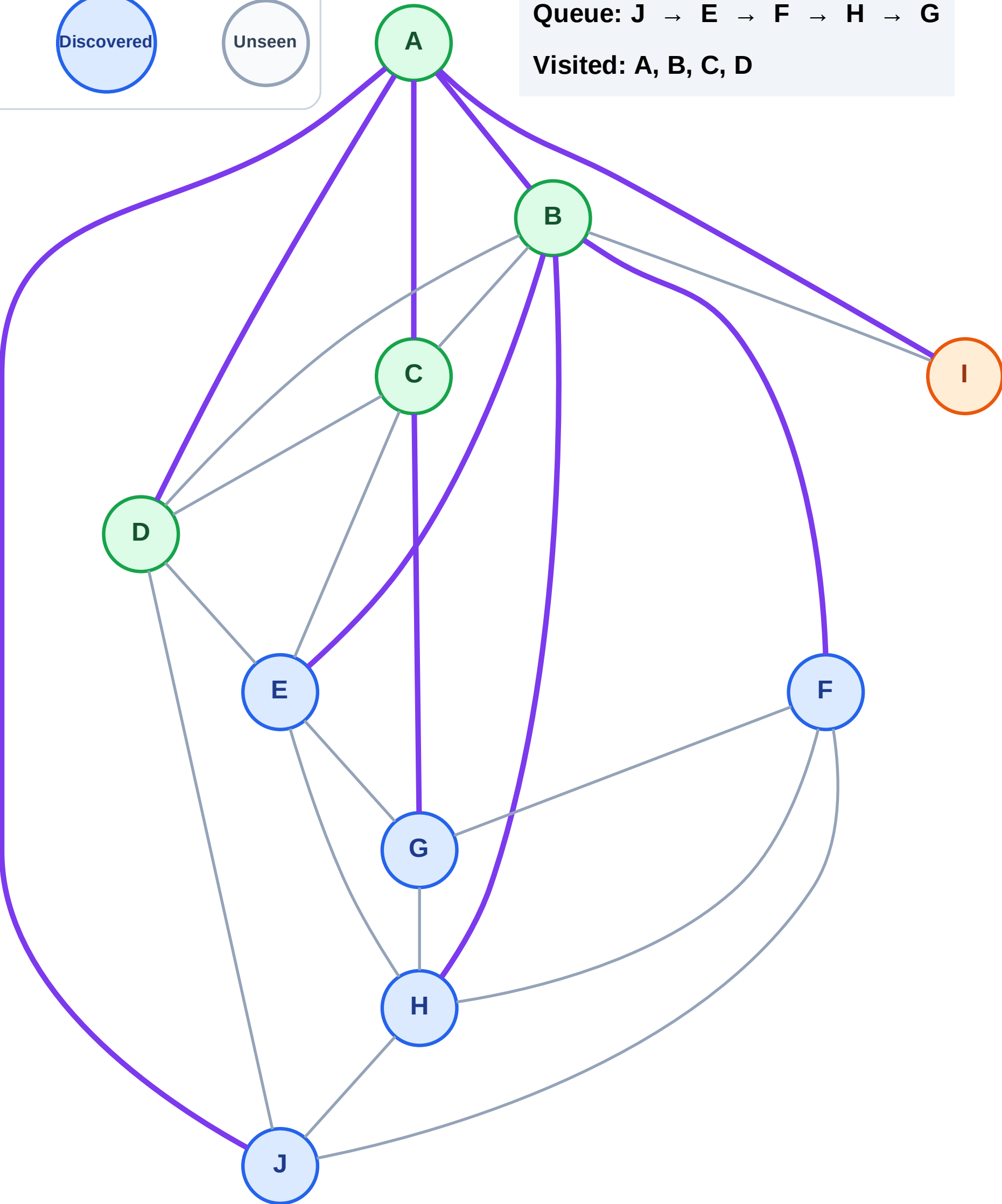
Processing

Discovered

Unseen

Queue: J → E → F → H → G

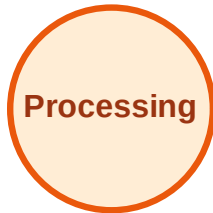
Visited: A, B, C, D



# BFS Traversal

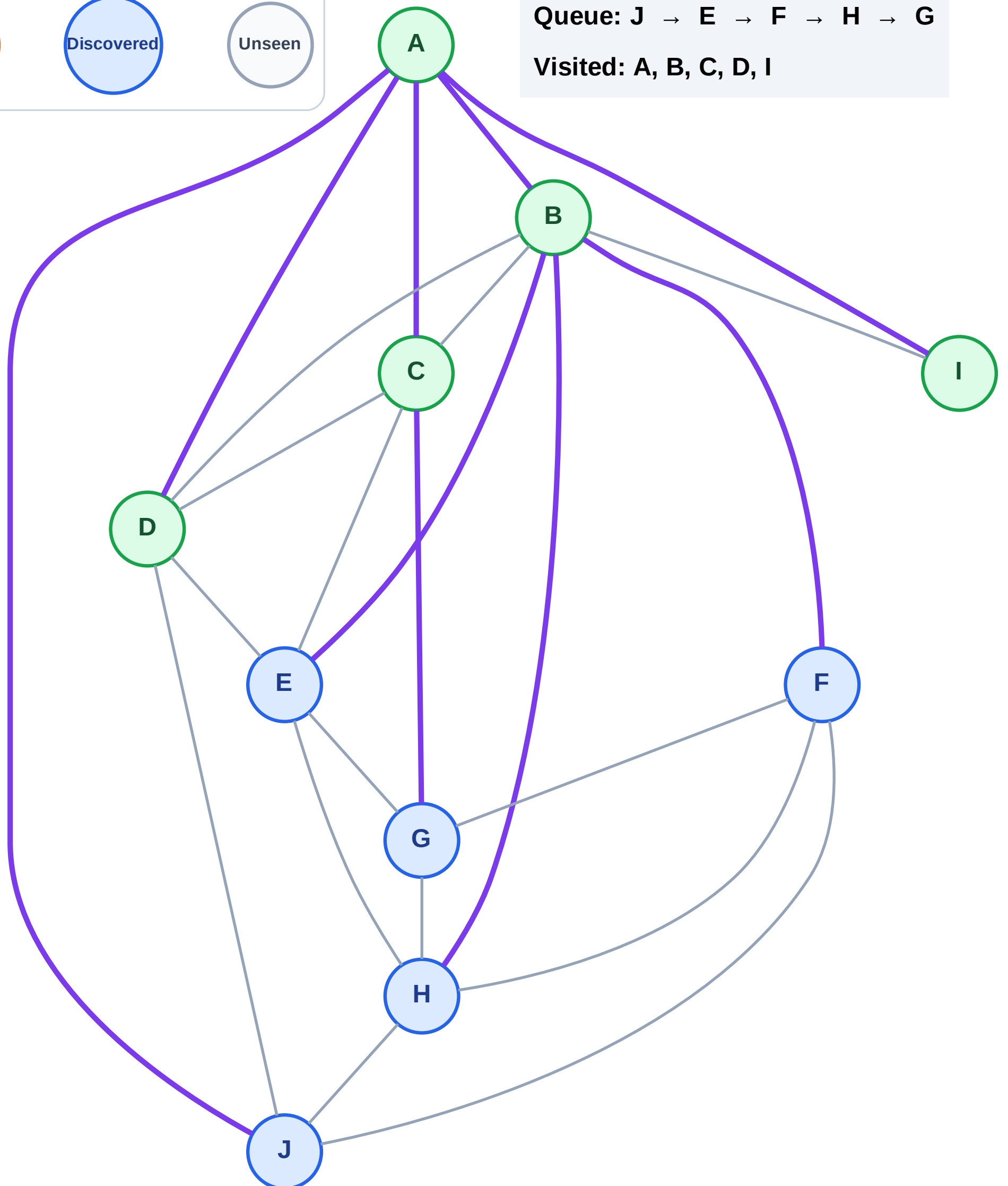
Step 11: No new neighbors from I

## Legend



Queue: J → E → F → H → G

Visited: A, B, C, D, I



# BFS Traversal

Step 12: Process node J

Legend

Complete

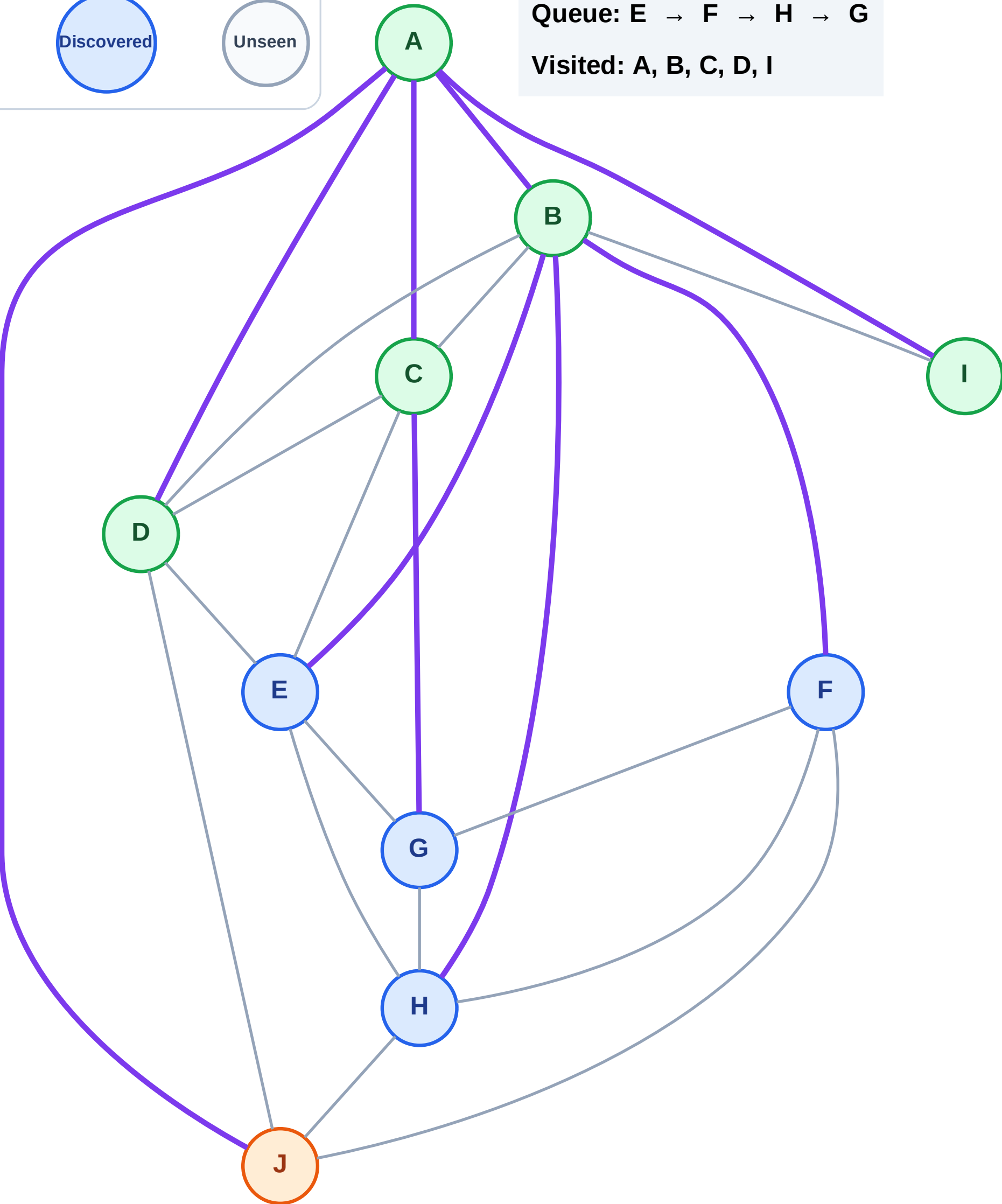
Processing

Discovered

Unseen

Queue: E → F → H → G

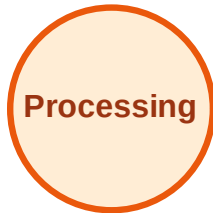
Visited: A, B, C, D, I



# BFS Traversal

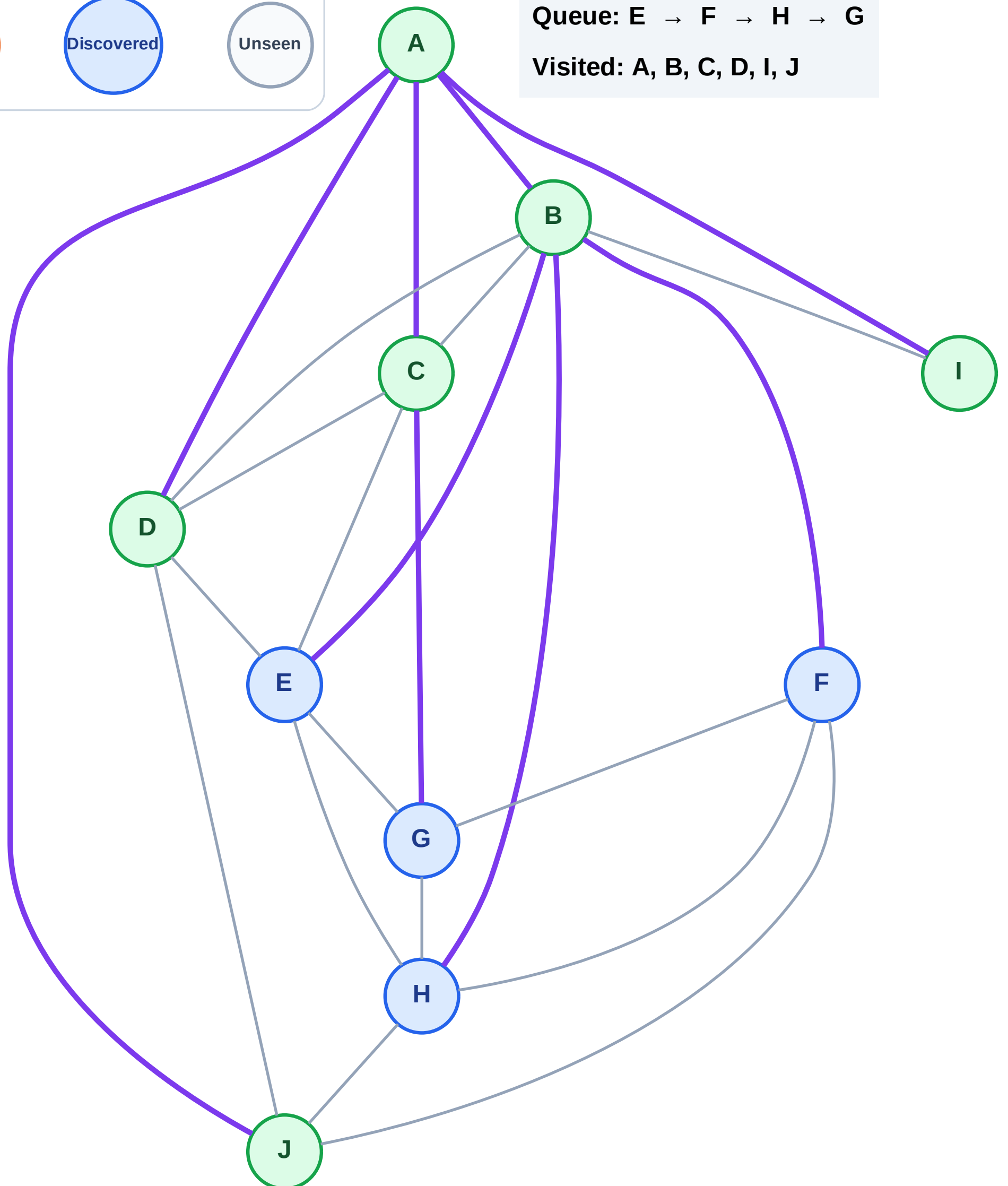
Step 13: No new neighbors from J

## Legend



Queue: E → F → H → G

Visited: A, B, C, D, I, J



# BFS Traversal

Step 14: Process node E

Legend

Complete

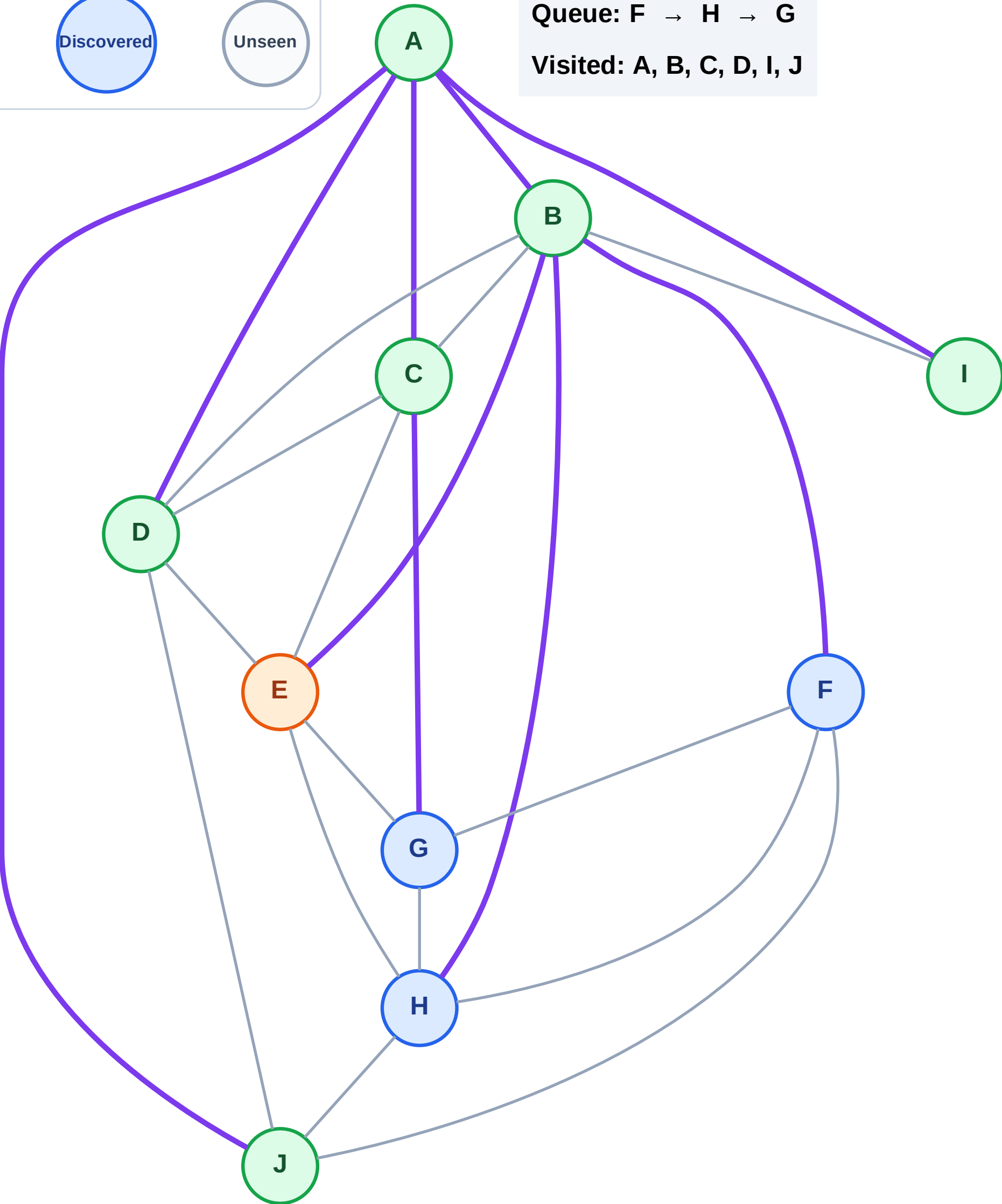
Processing

Discovered

Unseen

Queue: F → H → G

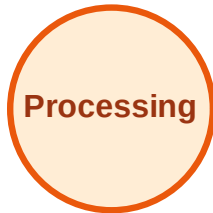
Visited: A, B, C, D, I, J



# BFS Traversal

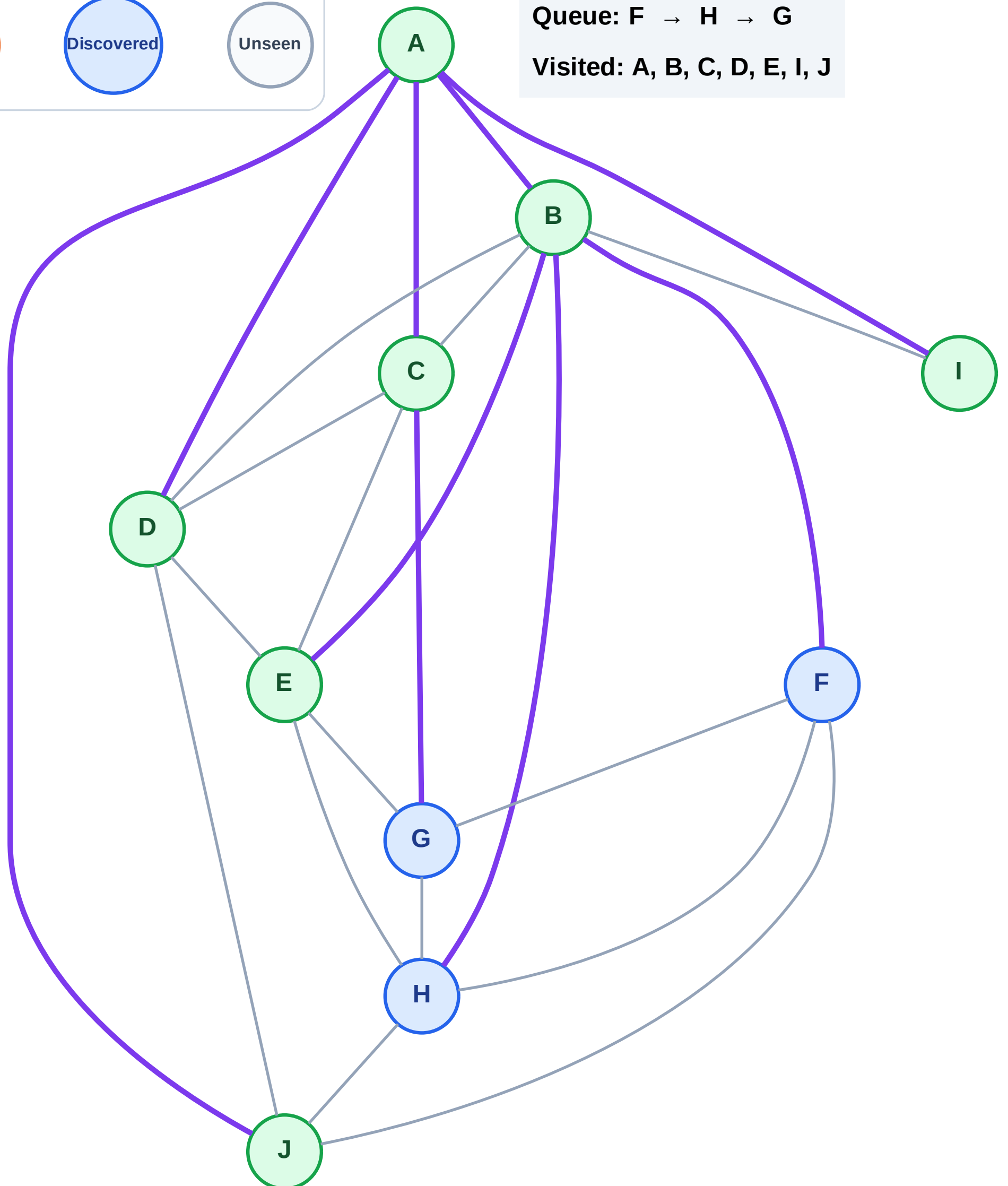
Step 15: No new neighbors from E

## Legend



Queue: F → H → G

Visited: A, B, C, D, E, I, J



# BFS Traversal

Step 16: Process node F

Legend

Complete

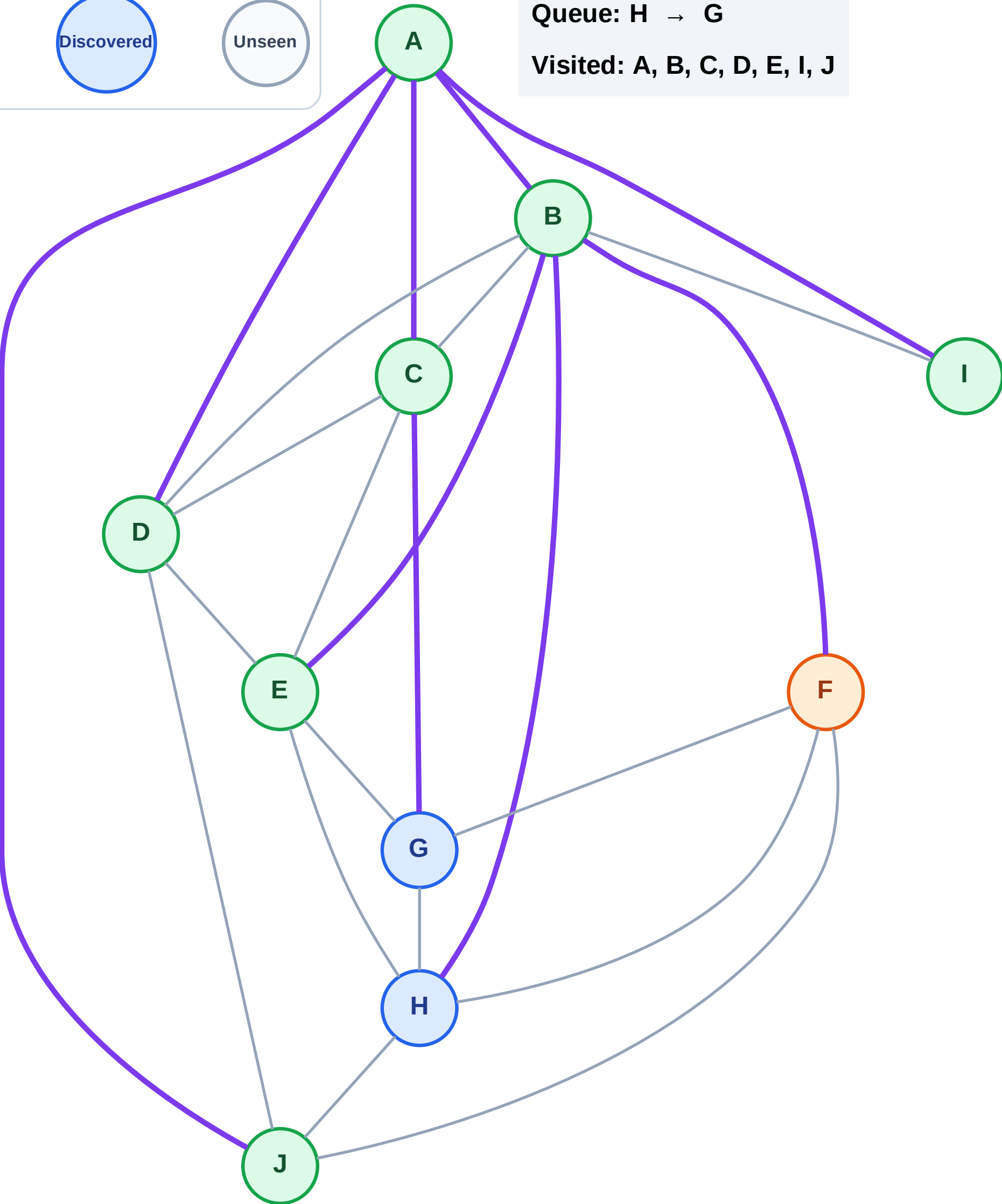
Processing

Discovered

Unseen

Queue: H → G

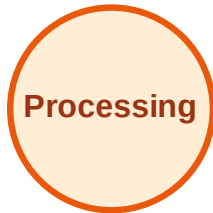
Visited: A, B, C, D, E, I, J



# BFS Traversal

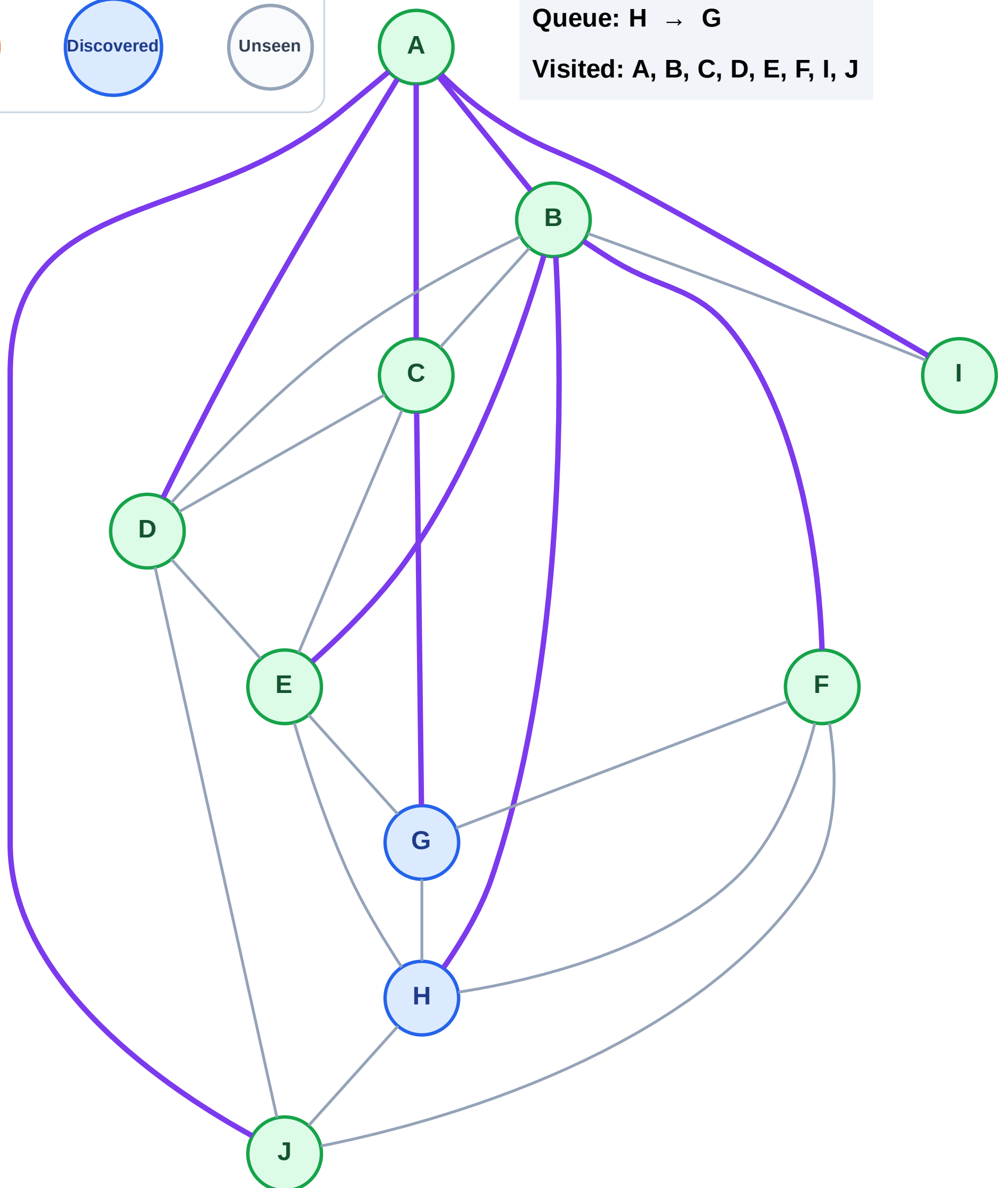
Step 17: No new neighbors from F

## Legend



Queue: H → G

Visited: A, B, C, D, E, F, I, J



# BFS Traversal

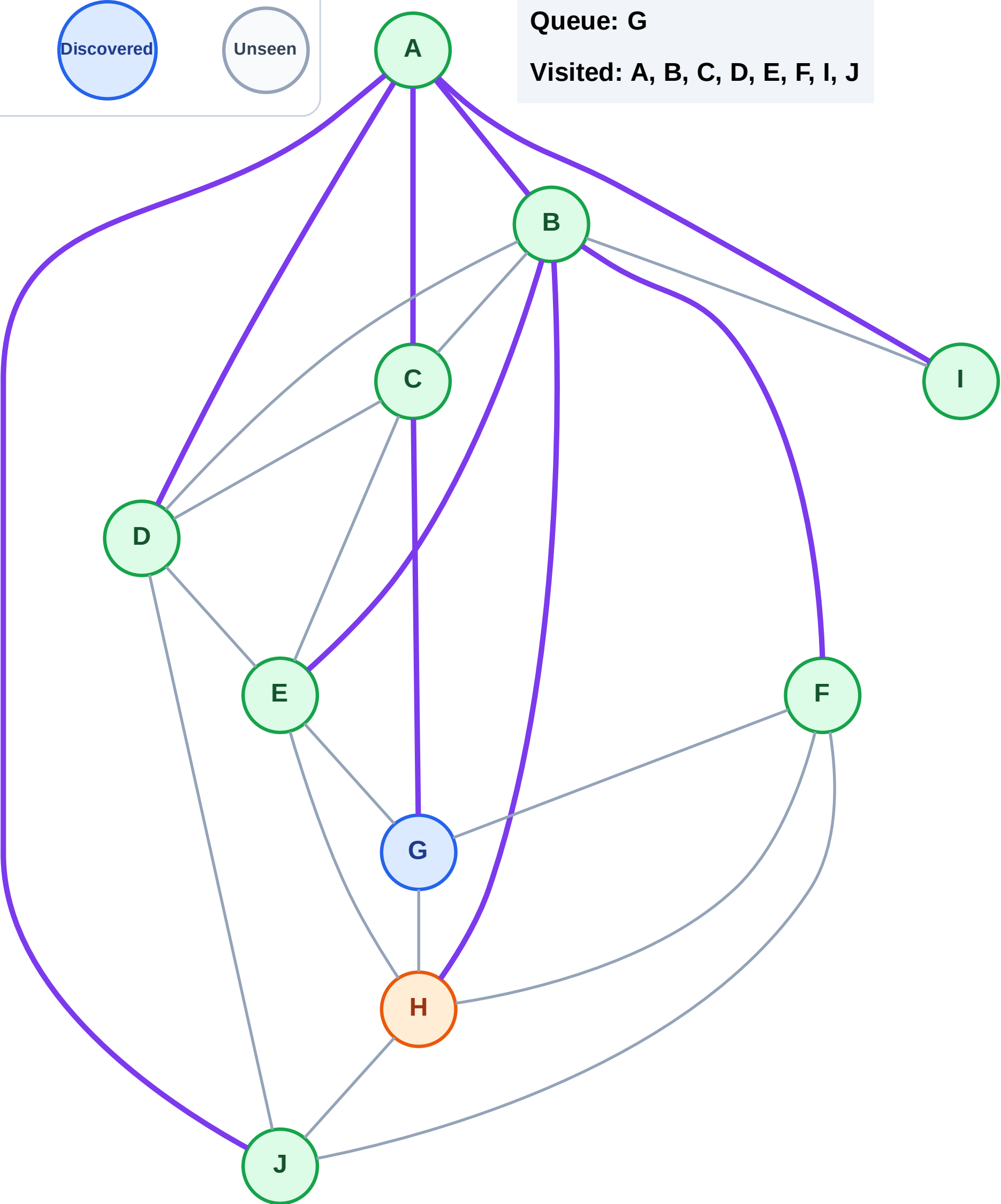
Step 18: Process node H

## Legend



Queue: G

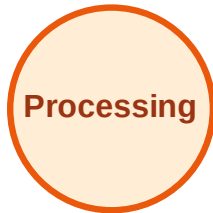
Visited: A, B, C, D, E, F, I, J



# BFS Traversal

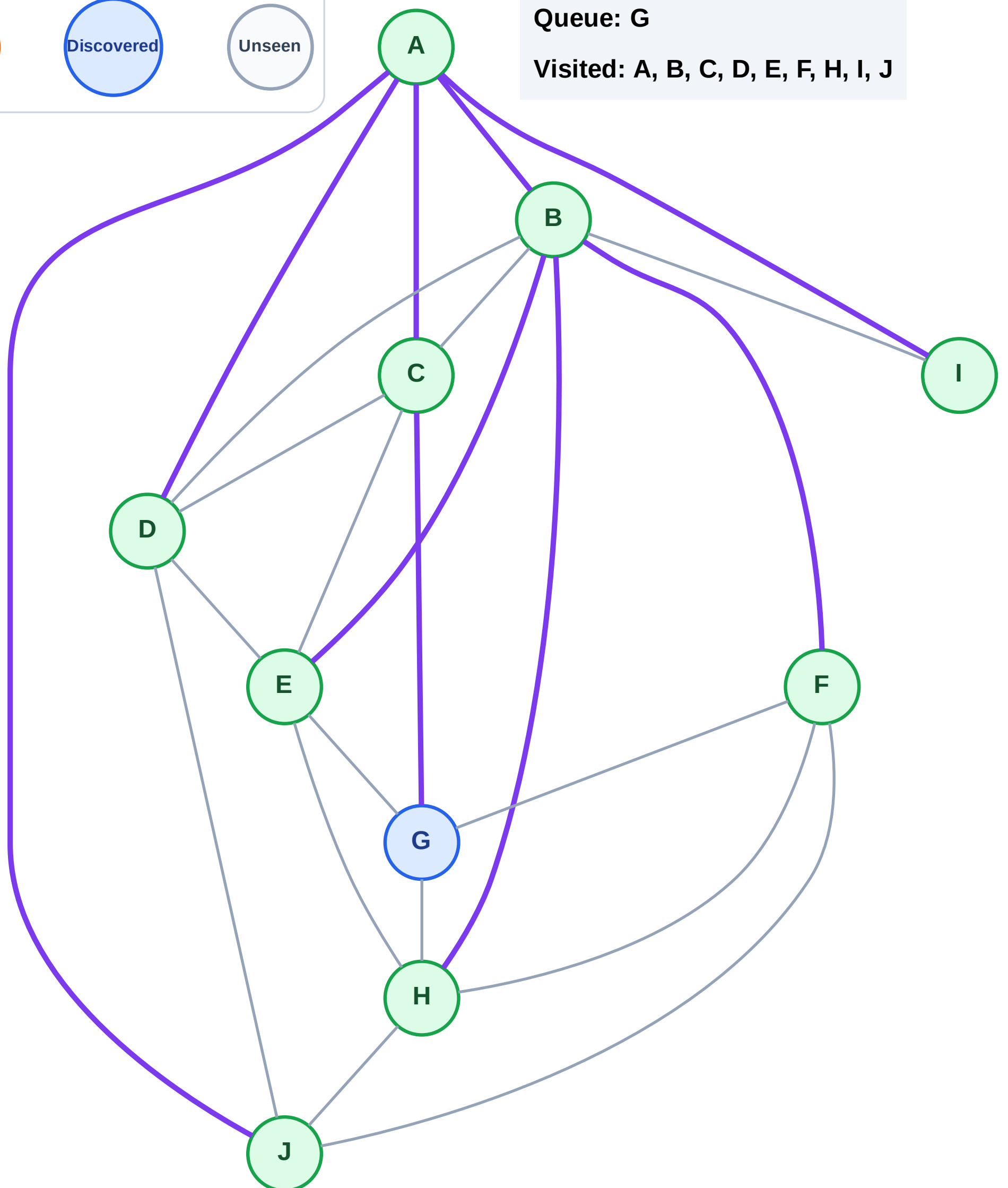
Step 19: No new neighbors from H

## Legend



Queue: G

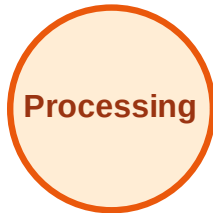
Visited: A, B, C, D, E, F, H, I, J



# BFS Traversal

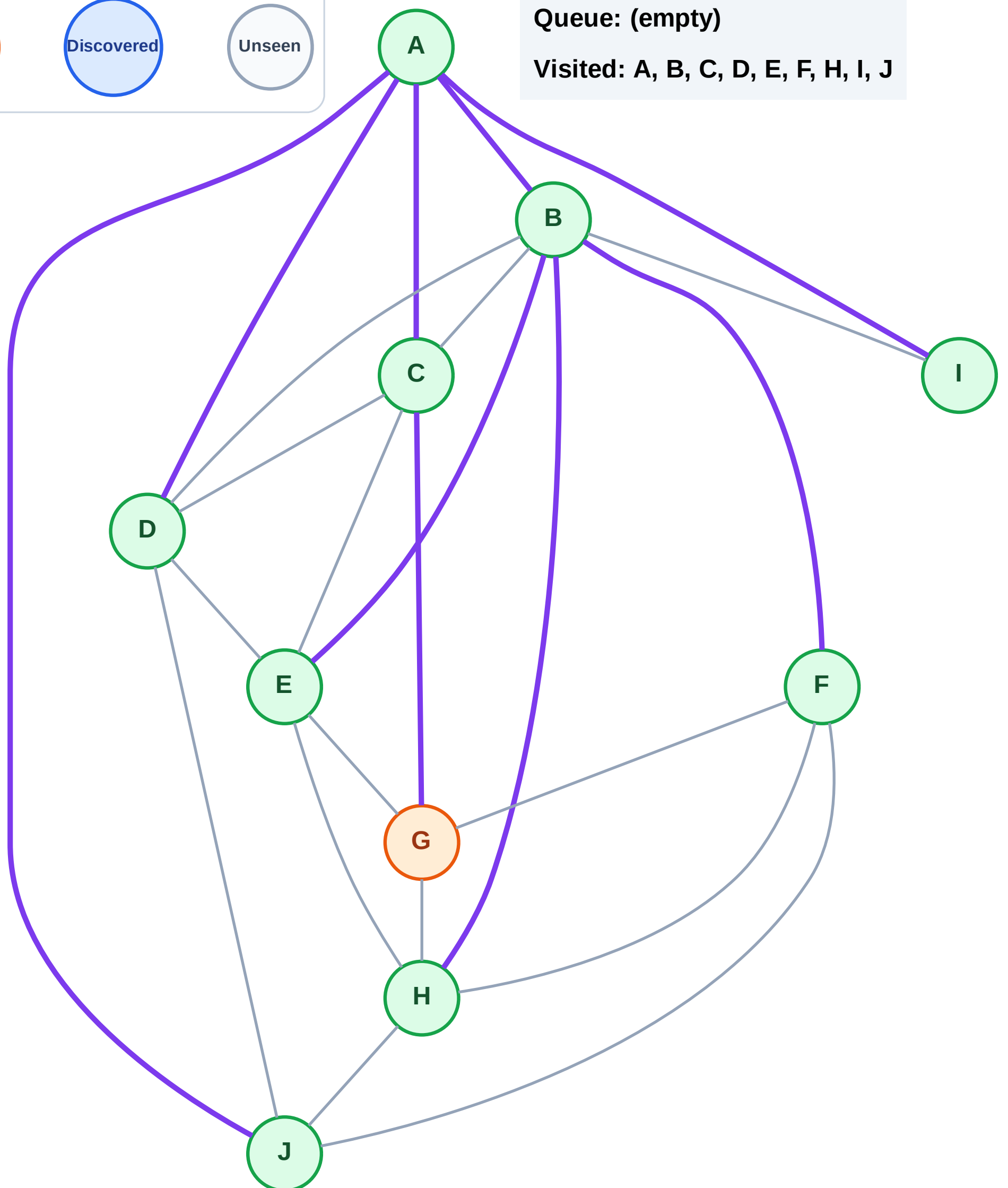
Step 20: Process node G

## Legend



Queue: (empty)

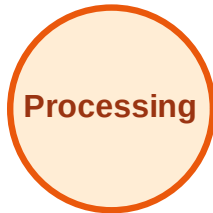
Visited: A, B, C, D, E, F, H, I, J



# BFS Traversal

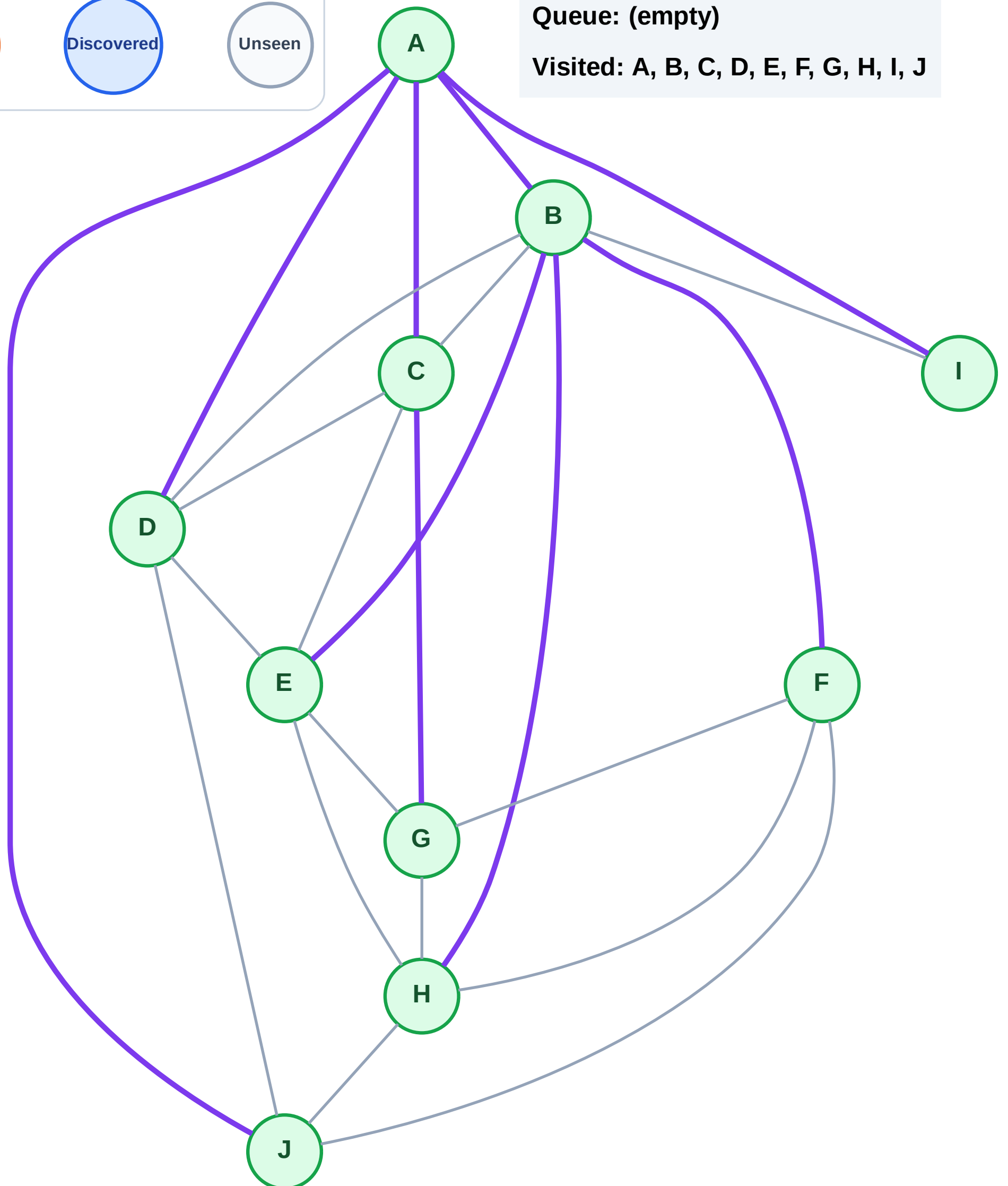
Step 21: No new neighbors from G

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

Step 1: Start at A

Legend

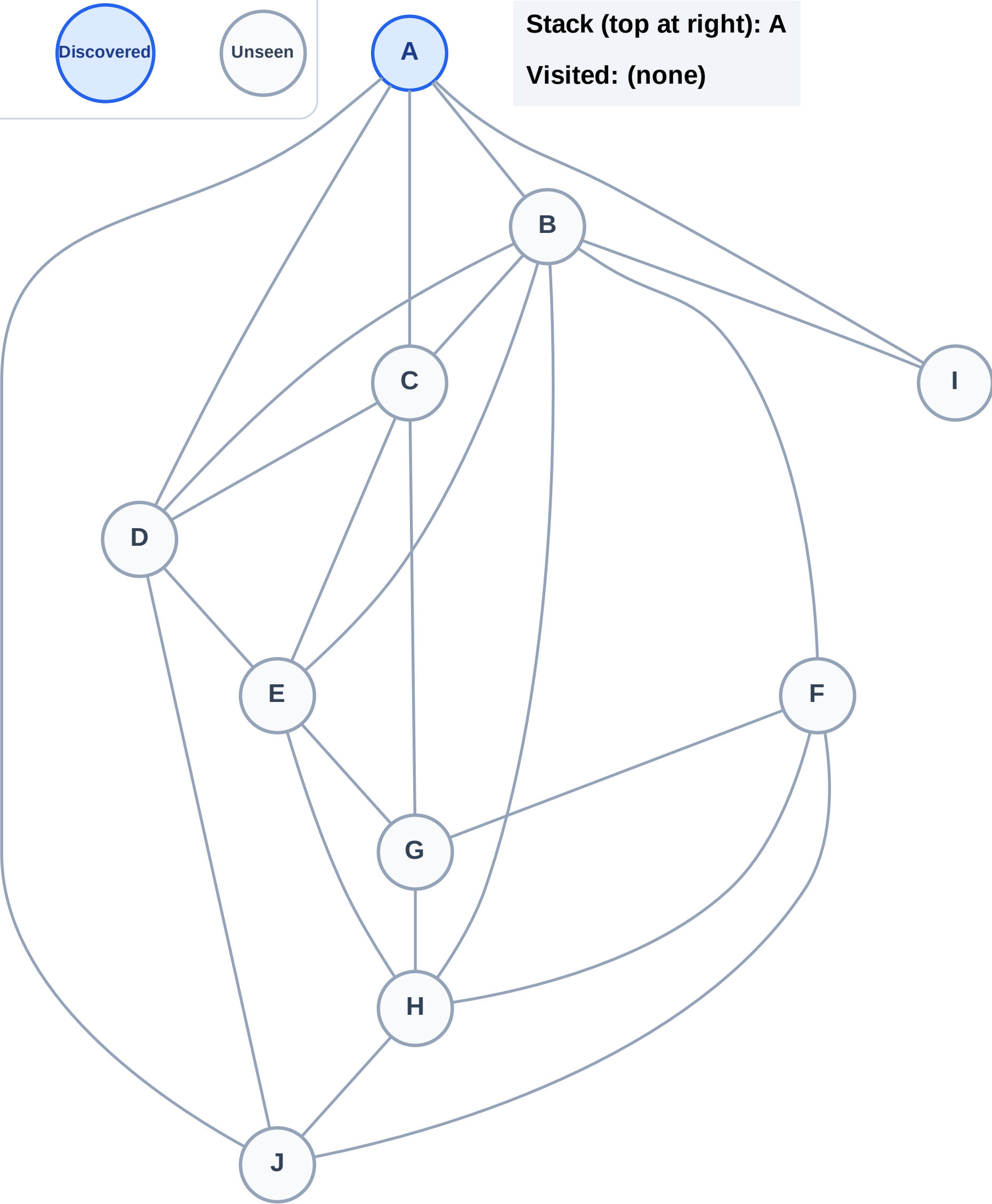
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

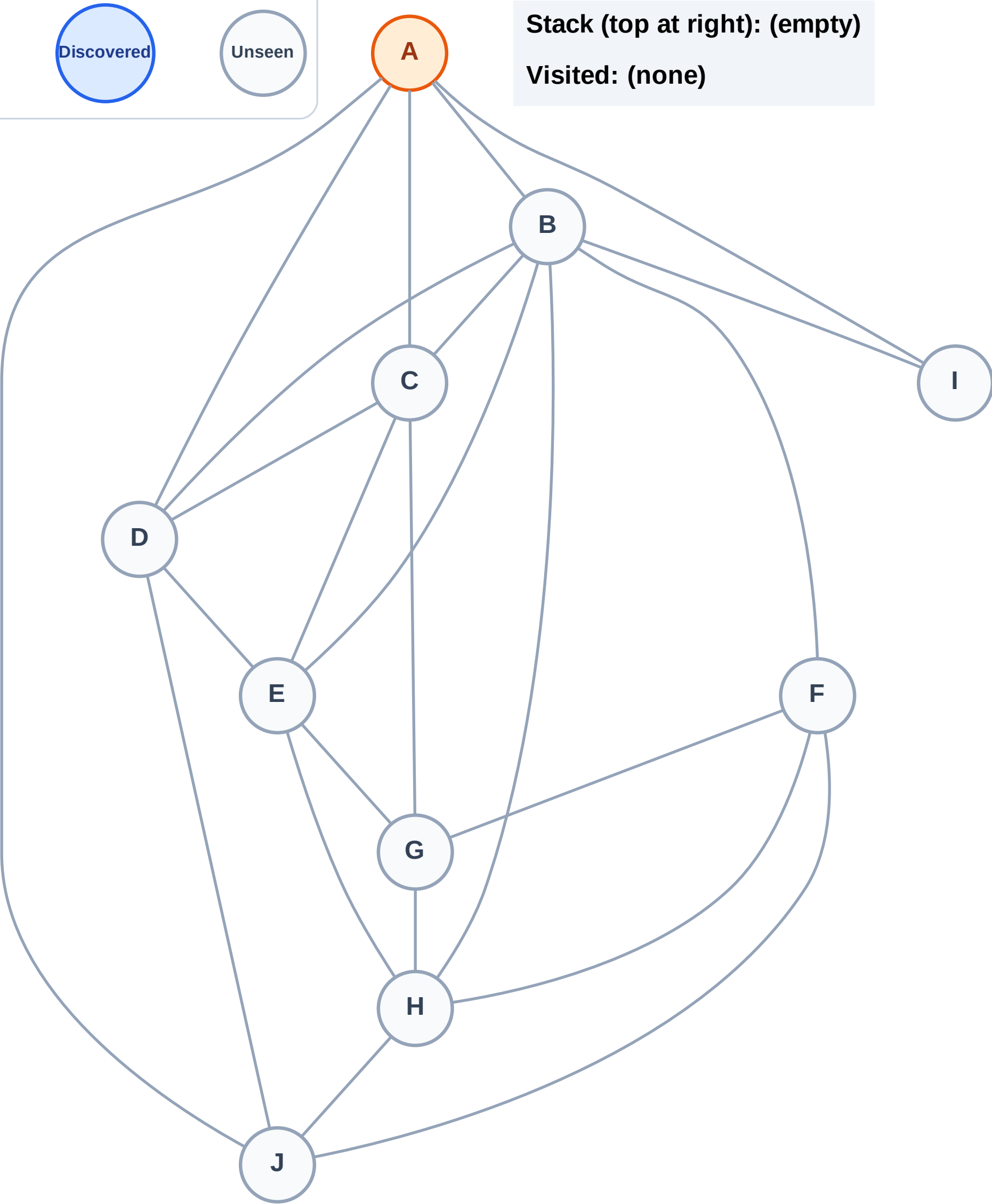
Complete

Processing

Discovered

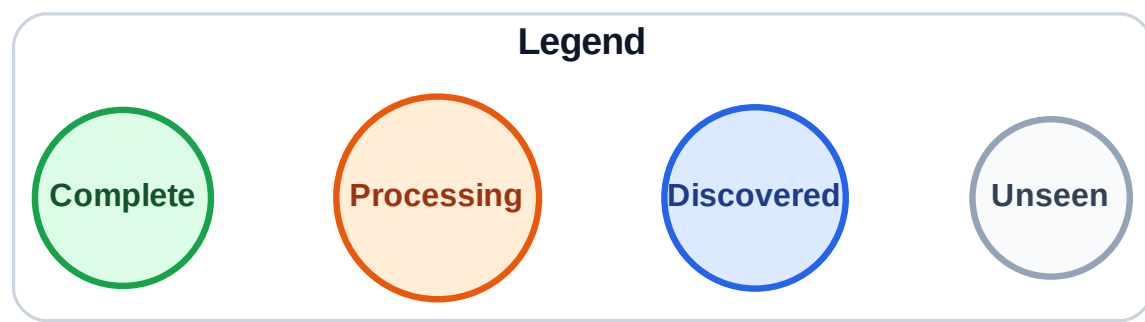
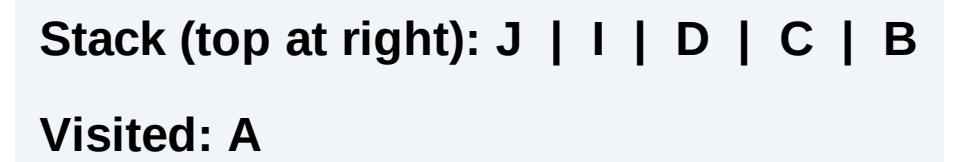
Unseen

Stack (top at right): (empty)  
Visited: (none)



### Step 3: Discover J, I, D, C, B from A

### Step 3: Discover J, I, D, C, B from A



# DFS Traversal

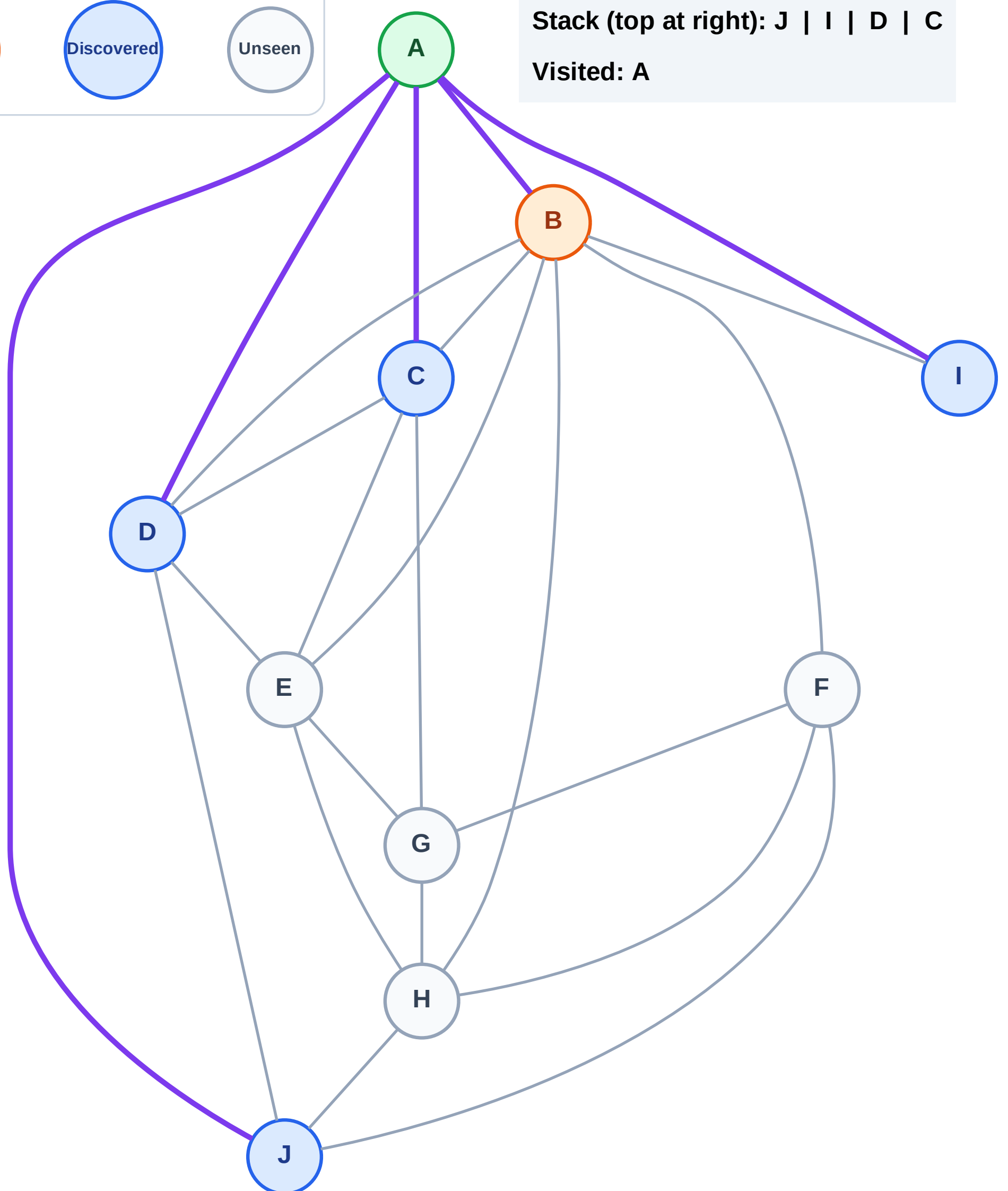
Step 4: Process node B

## Legend



Stack (top at right): J | I | D | C

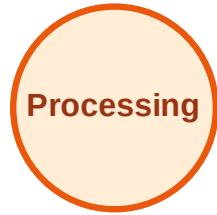
Visited: A



# DFS Traversal

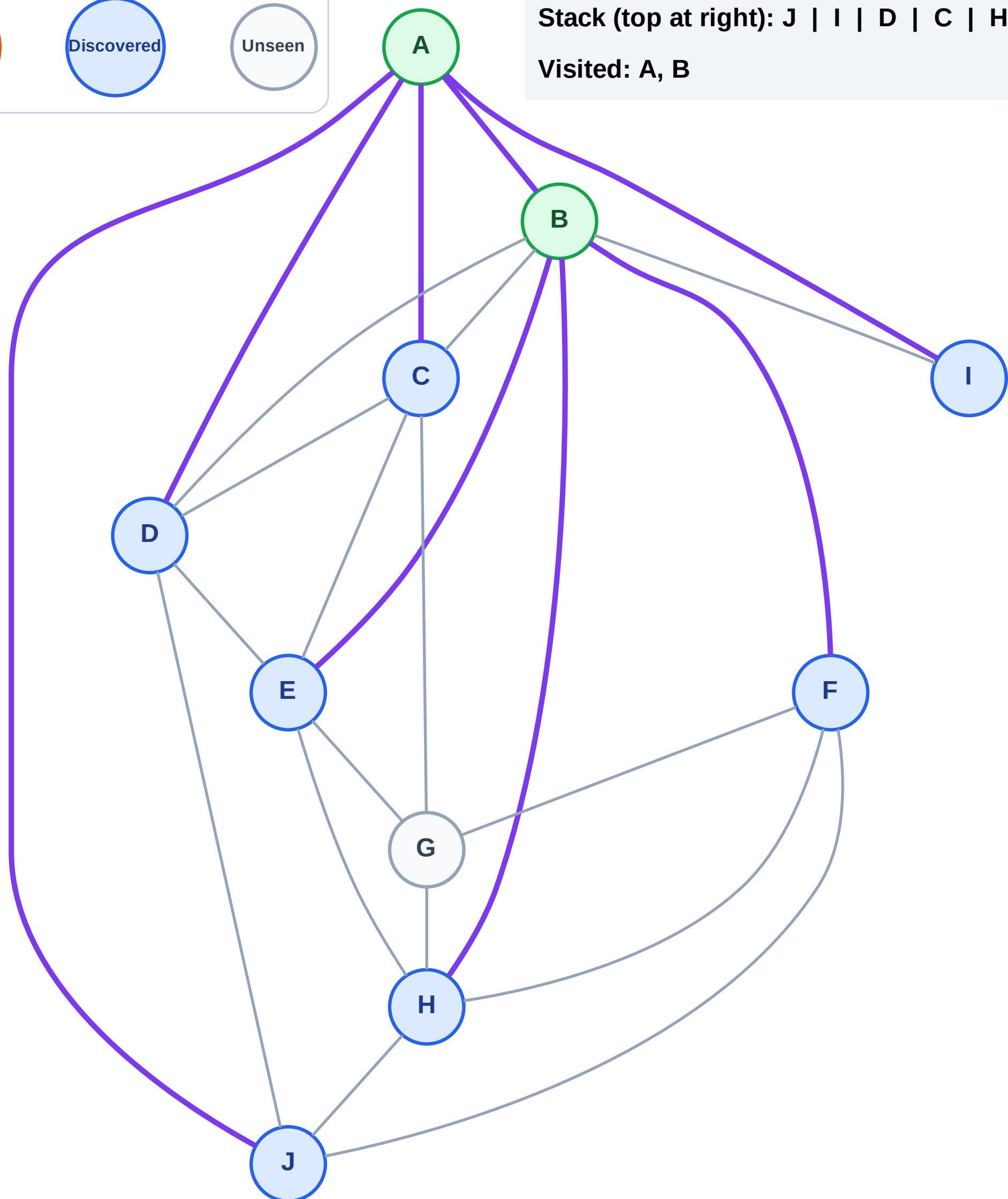
Step 5: Discover H, F, E from B

## Legend



Stack (top at right): J | I | D | C | H | F | E

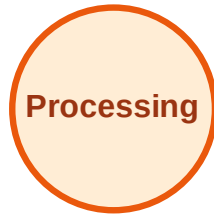
Visited: A, B



# DFS Traversal

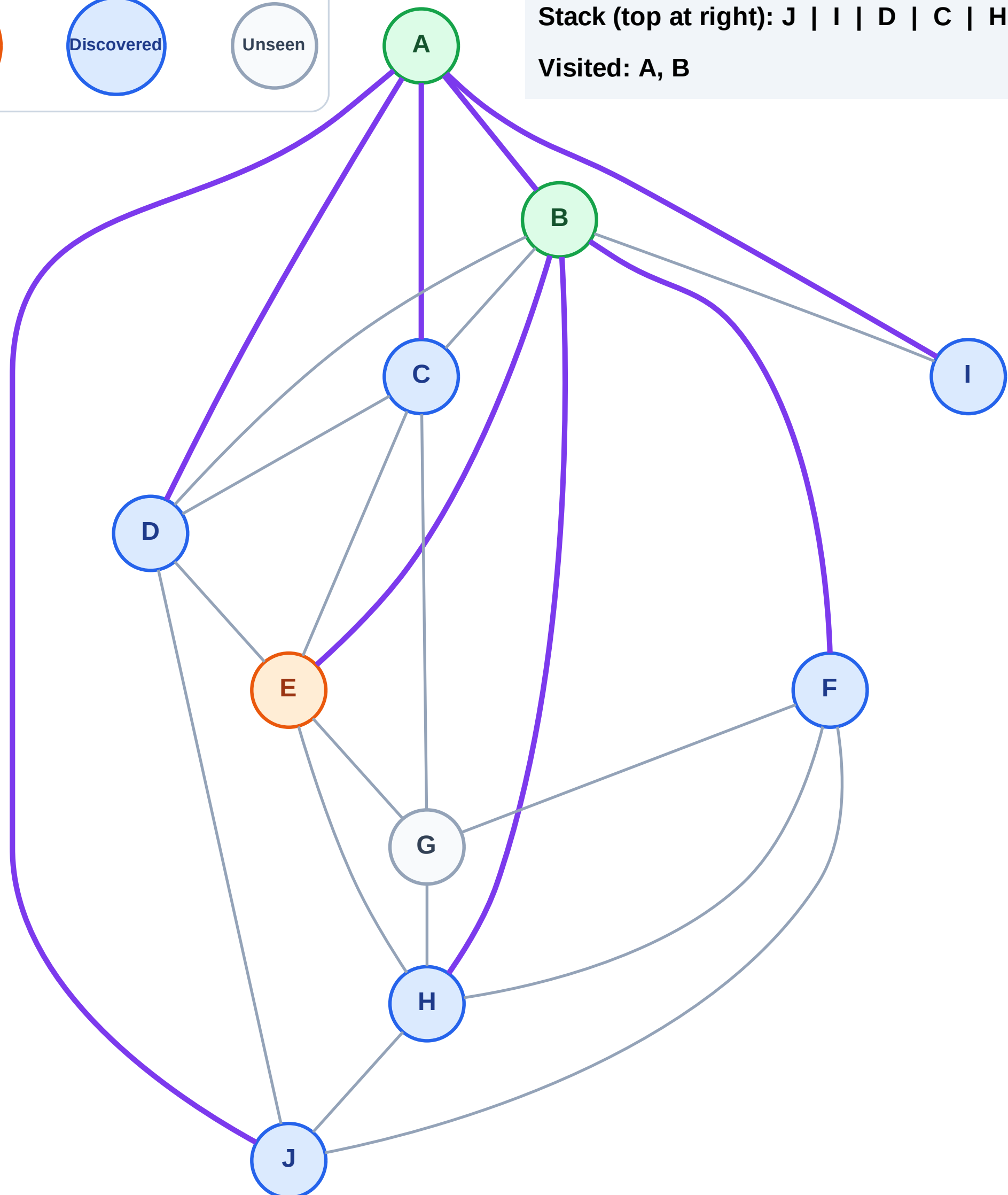
Step 6: Process node E

## Legend



Stack (top at right): J | I | D | C | H | F

Visited: A, B



# DFS Traversal

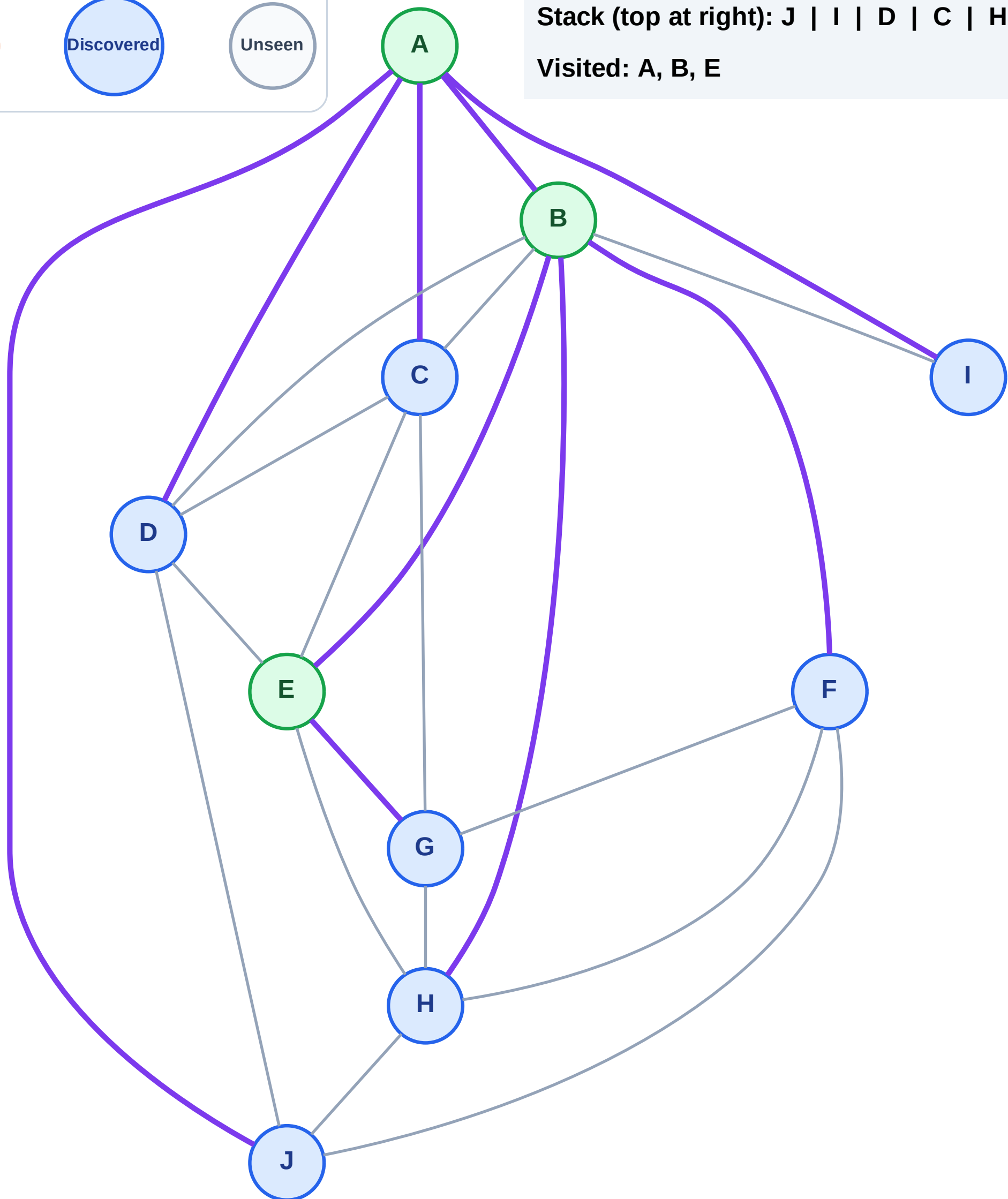
Step 7: Discover G from E

## Legend



Stack (top at right): J | I | D | C | H | F | G

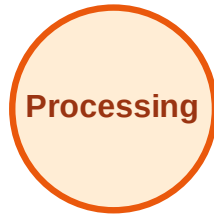
Visited: A, B, E



# DFS Traversal

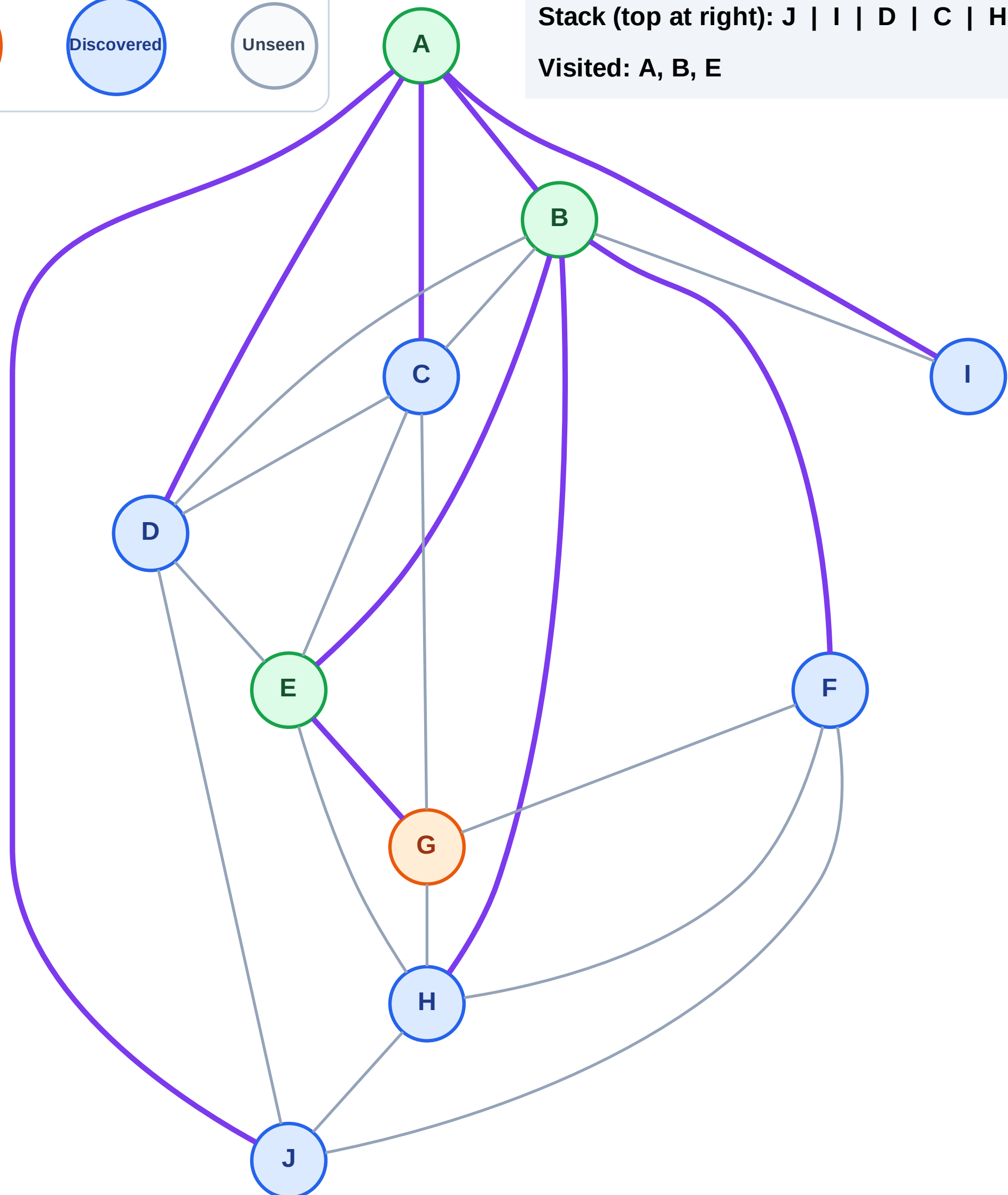
Step 8: Process node G

## Legend



Stack (top at right): J | I | D | C | H | F

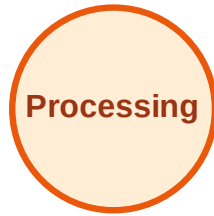
Visited: A, B, E



# DFS Traversal

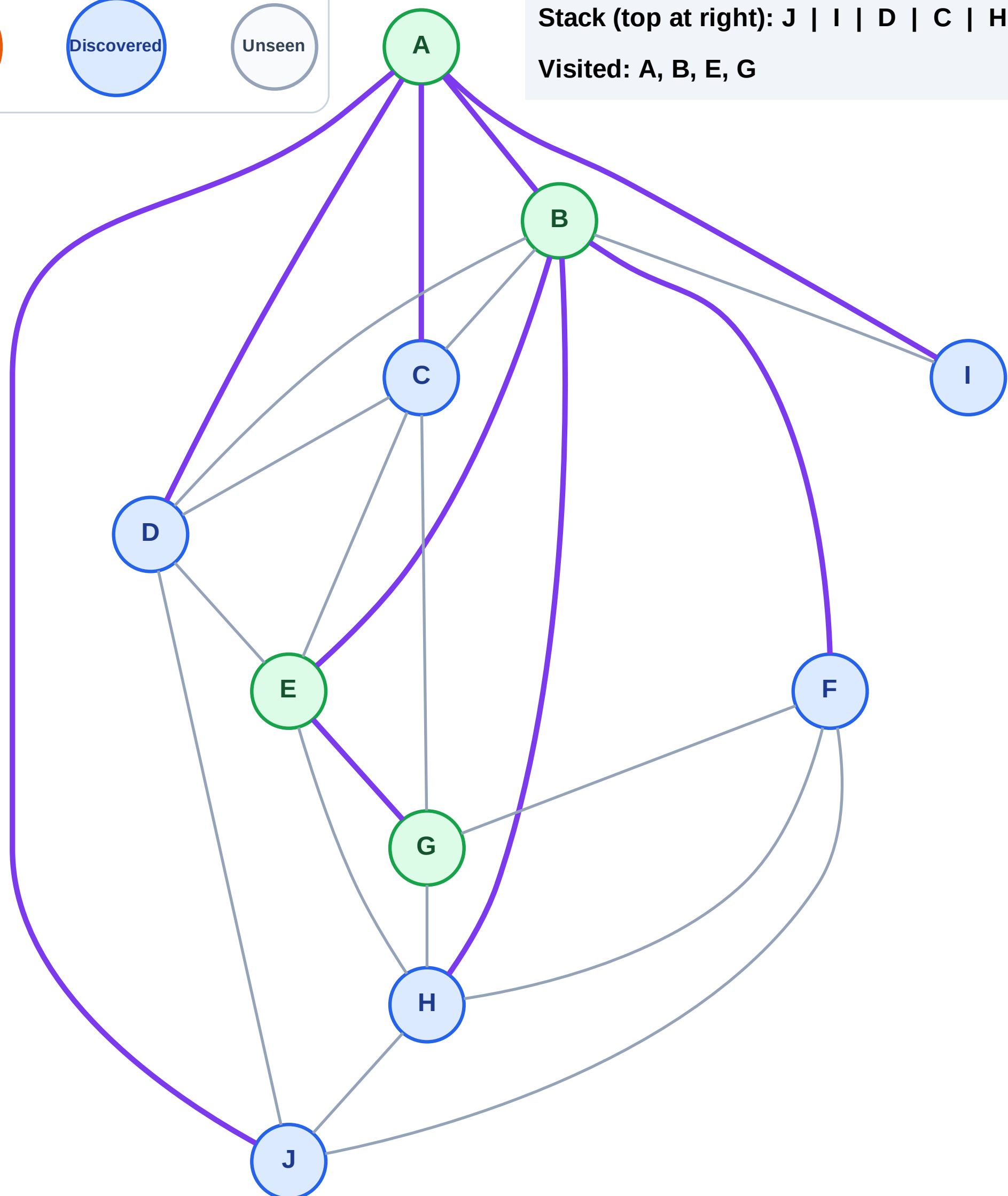
Step 9: No new neighbors from G

## Legend



Stack (top at right): J | I | D | C | H | F

Visited: A, B, E, G



# DFS Traversal

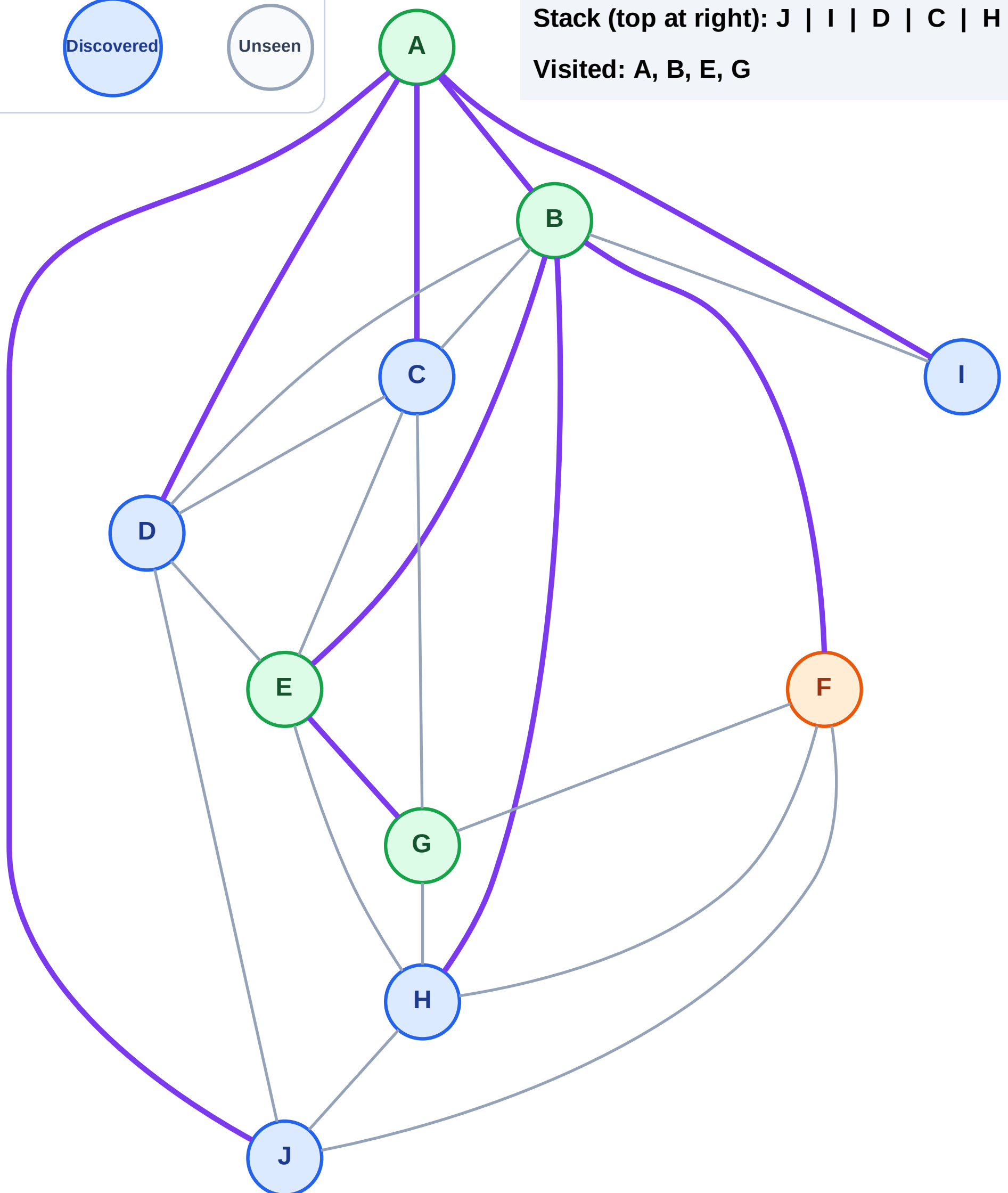
Step 10: Process node F

## Legend



Stack (top at right): J | I | D | C | H

Visited: A, B, E, G



# DFS Traversal

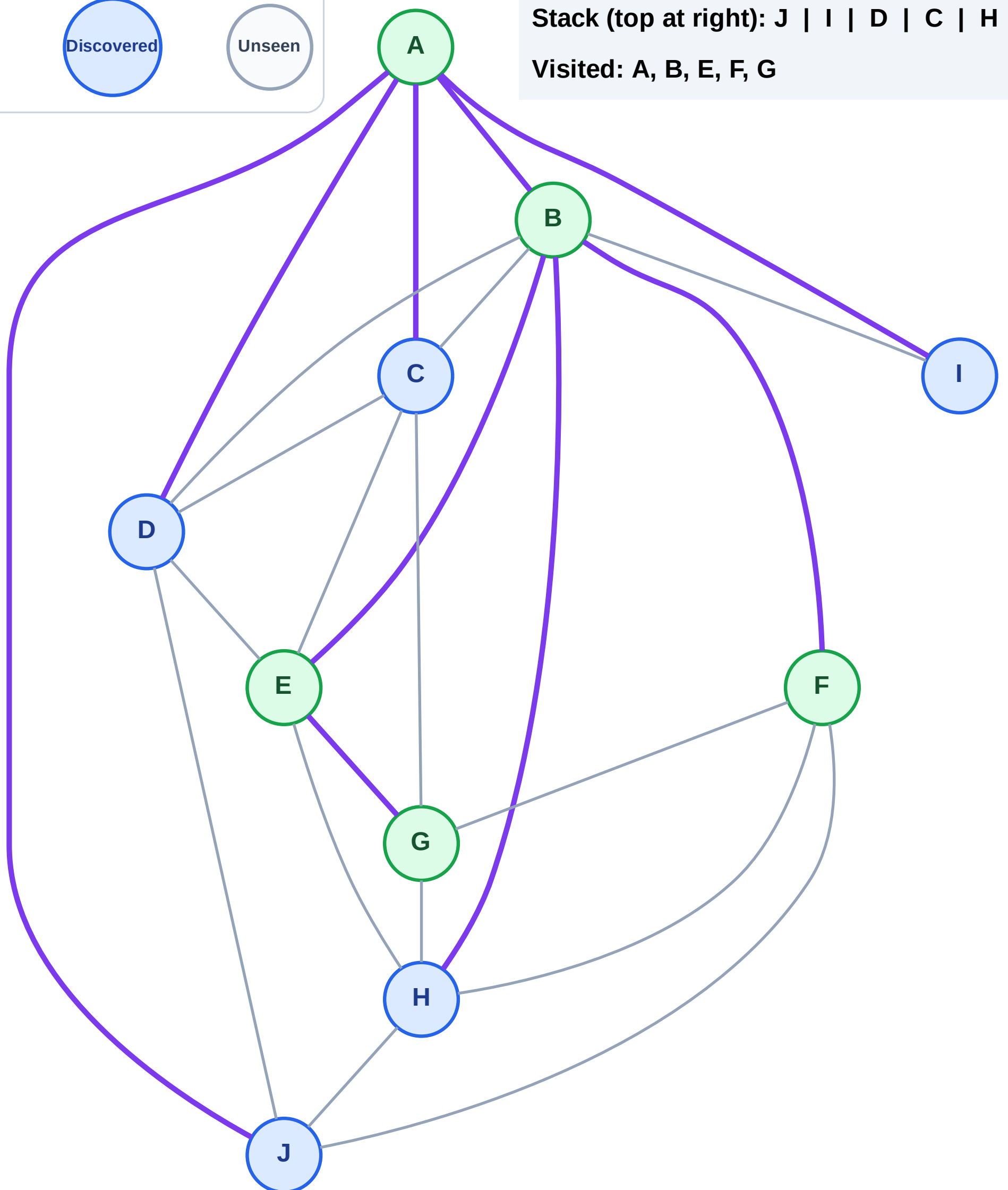
Step 11: No new neighbors from F

## Legend



Stack (top at right): J | I | D | C | H

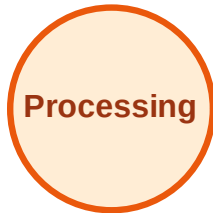
Visited: A, B, E, F, G



# DFS Traversal

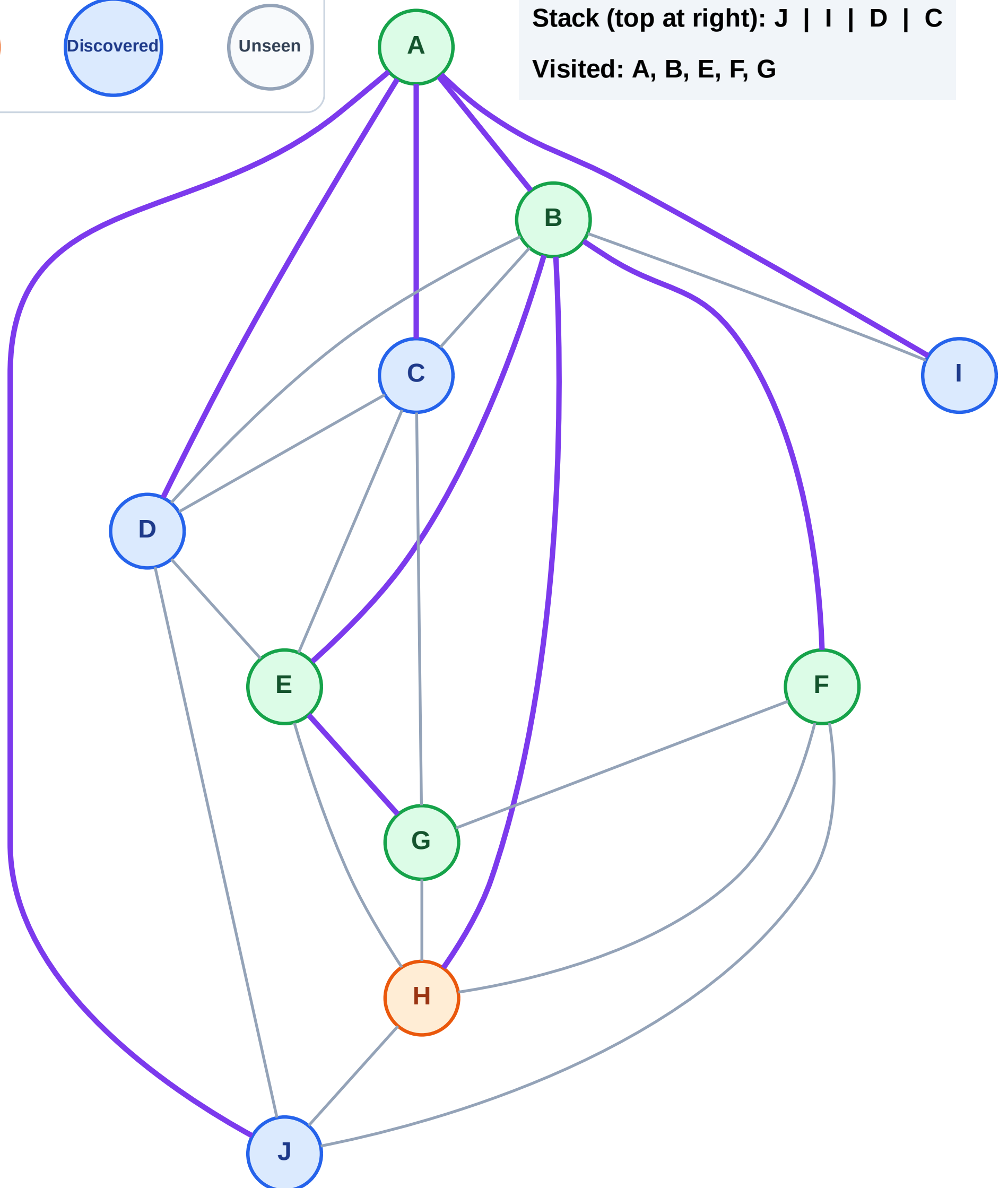
Step 12: Process node H

## Legend



Stack (top at right): J | I | D | C

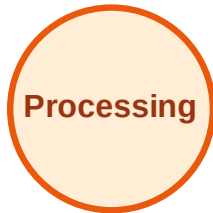
Visited: A, B, E, F, G



# DFS Traversal

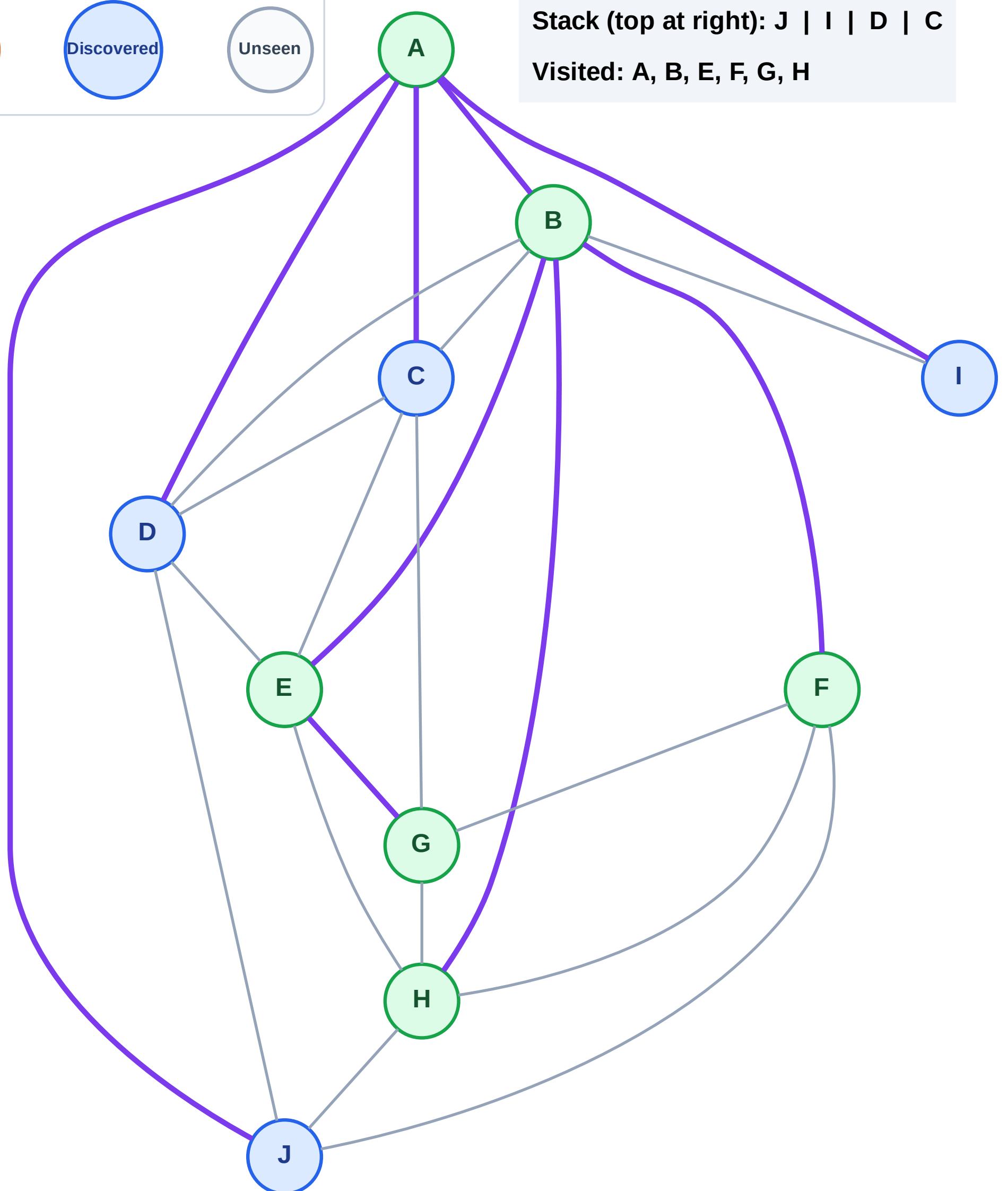
Step 13: No new neighbors from H

## Legend



Stack (top at right): J | I | D | C

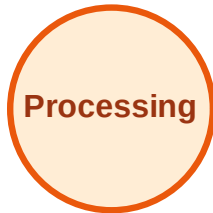
Visited: A, B, E, F, G, H



# DFS Traversal

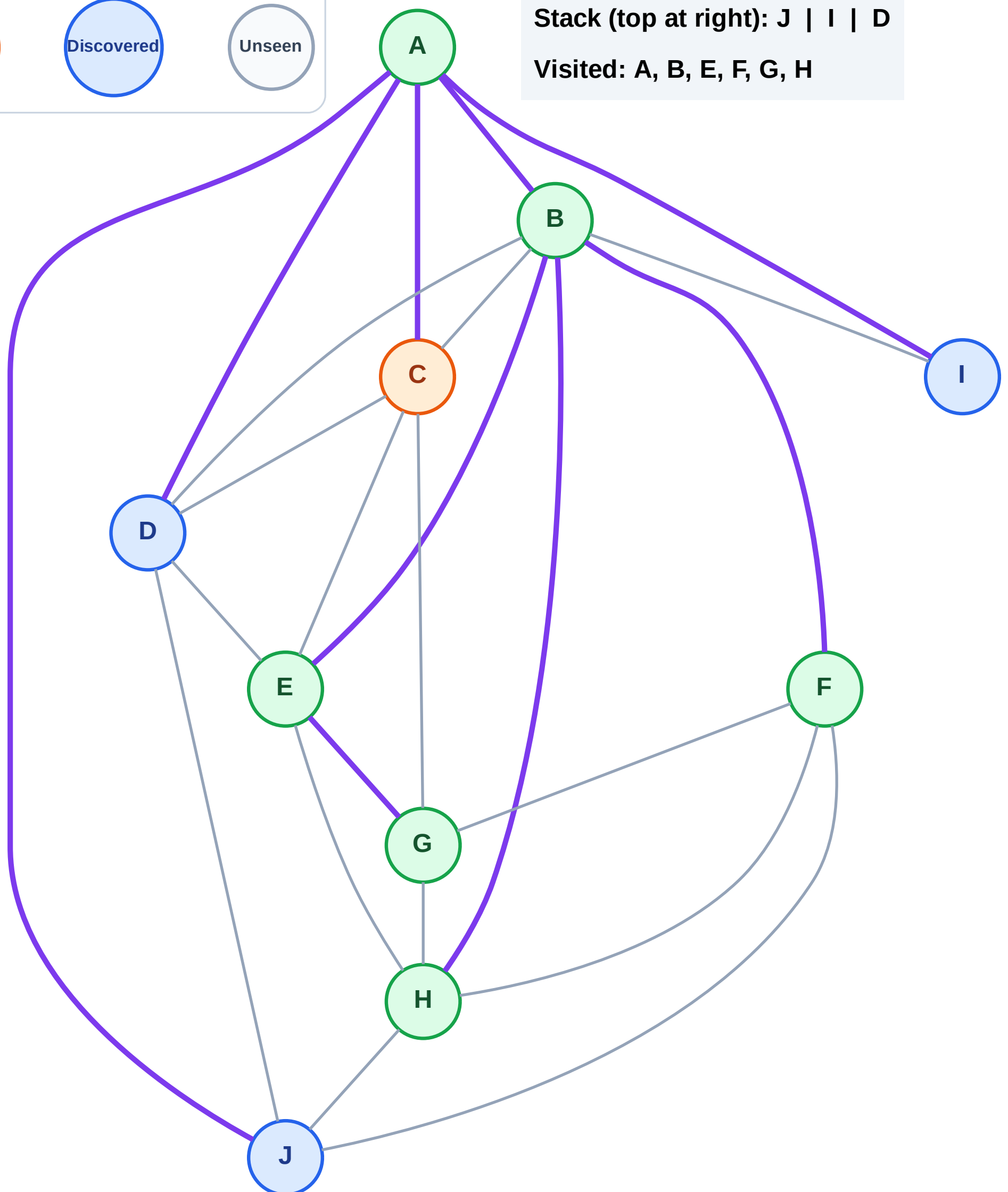
Step 14: Process node C

## Legend



Stack (top at right): J | I | D

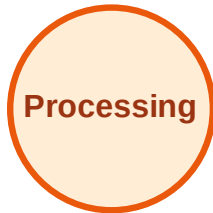
Visited: A, B, E, F, G, H



# DFS Traversal

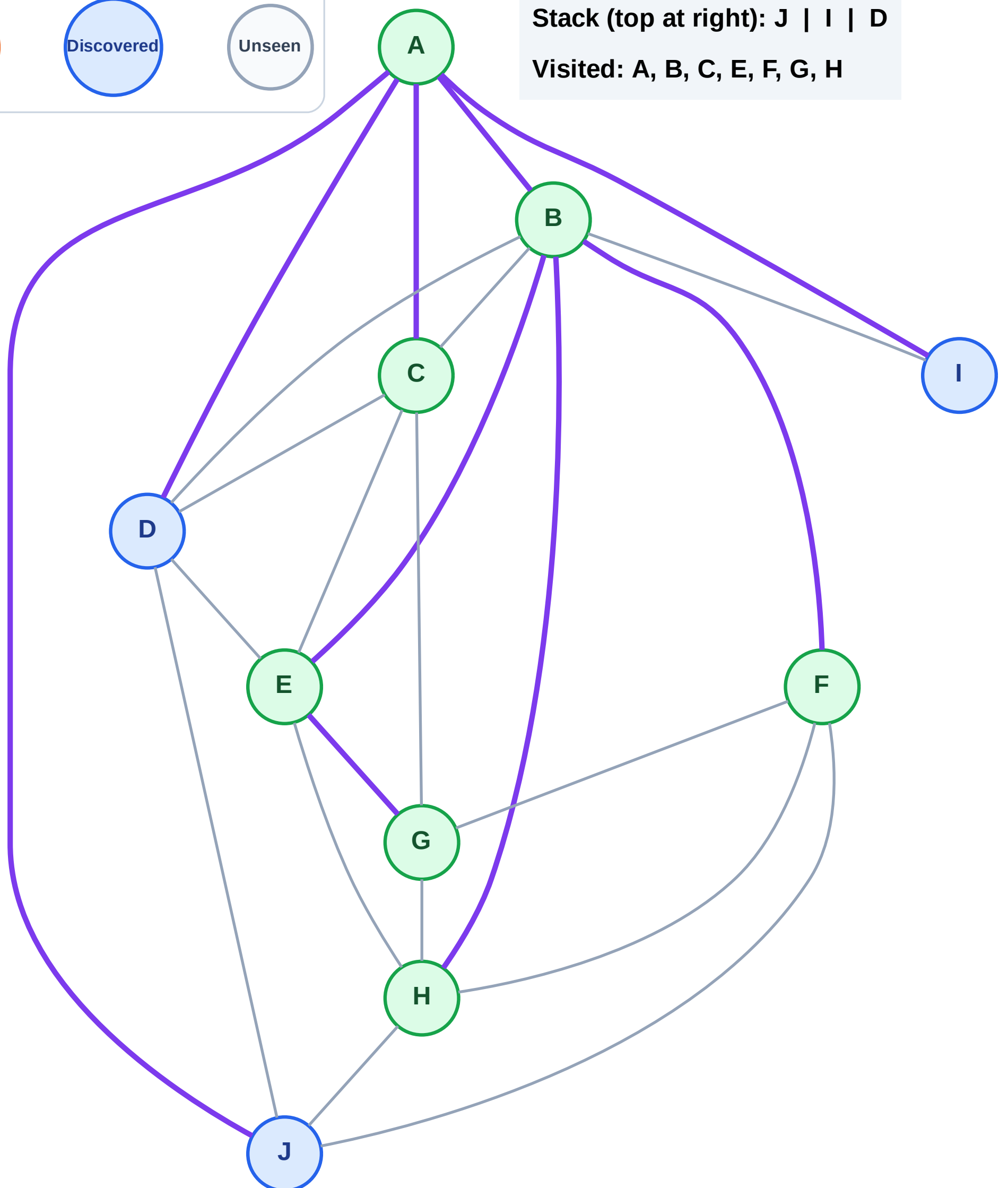
Step 15: No new neighbors from C

## Legend



Stack (top at right): J | I | D

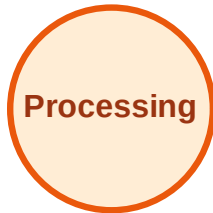
Visited: A, B, C, E, F, G, H



# DFS Traversal

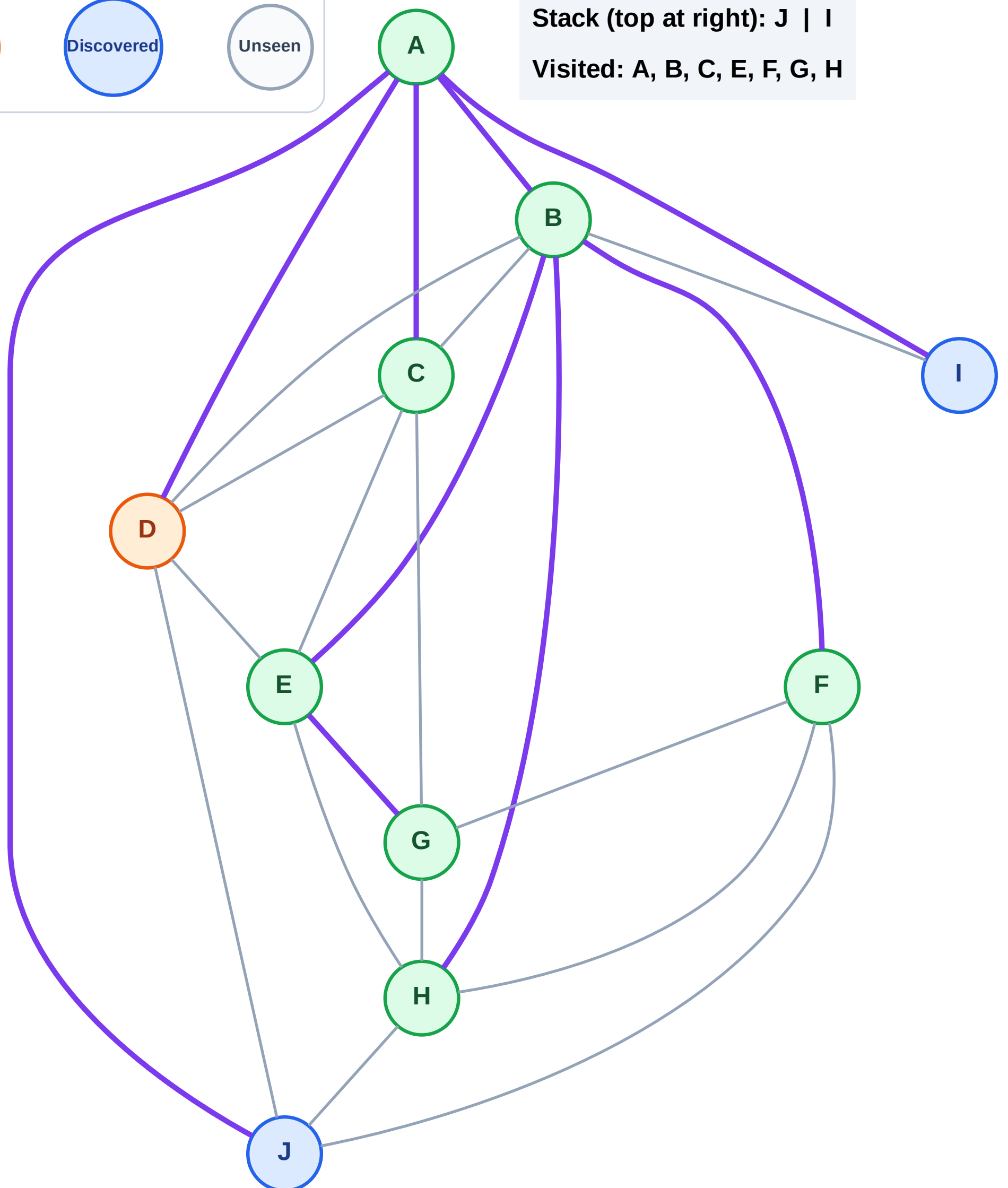
Step 16: Process node D

## Legend



Stack (top at right): J | I

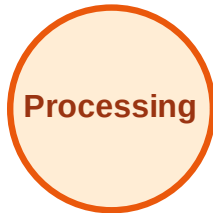
Visited: A, B, C, E, F, G, H



# DFS Traversal

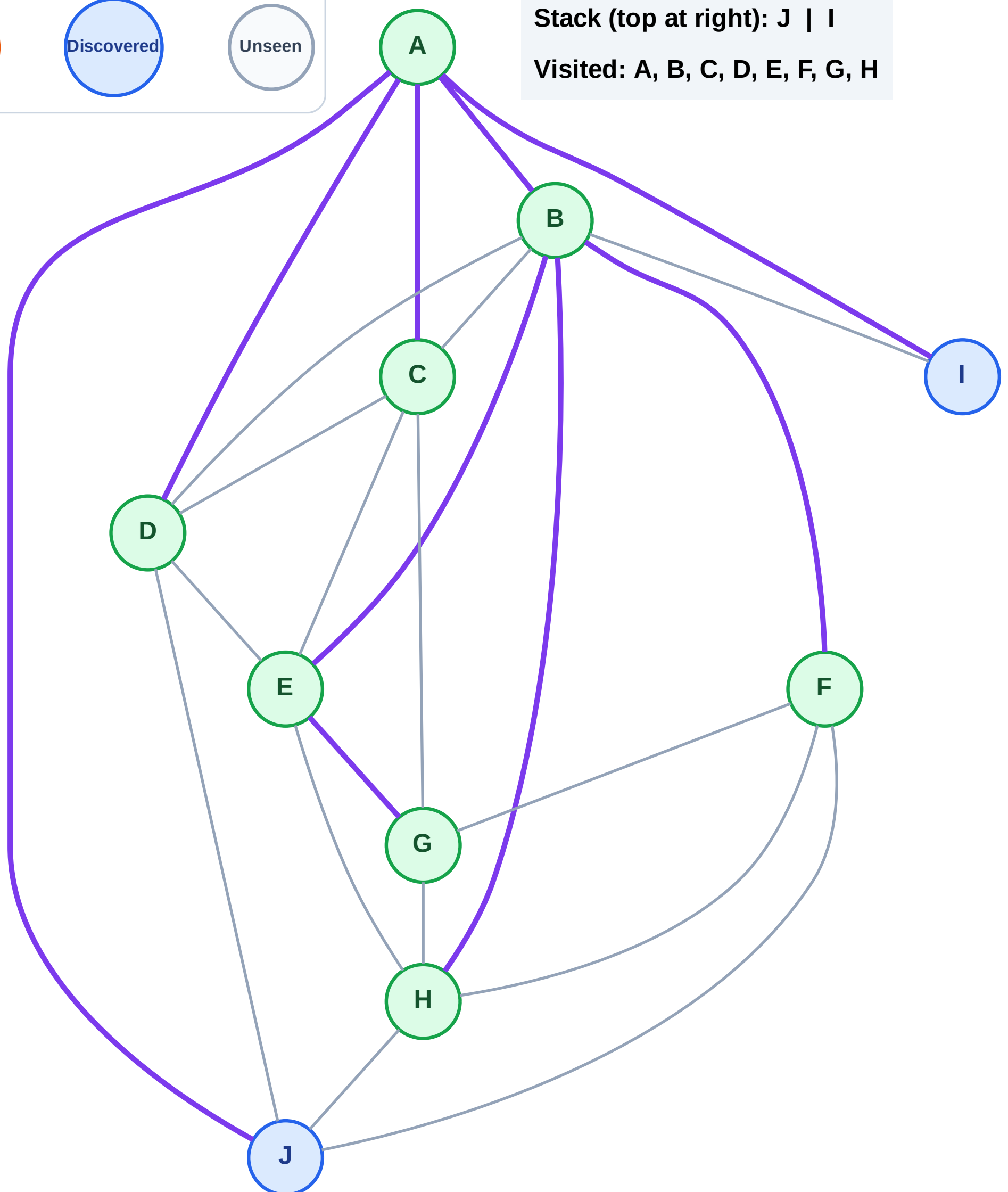
Step 17: No new neighbors from D

## Legend



Stack (top at right): J | I

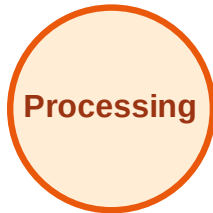
Visited: A, B, C, D, E, F, G, H



# DFS Traversal

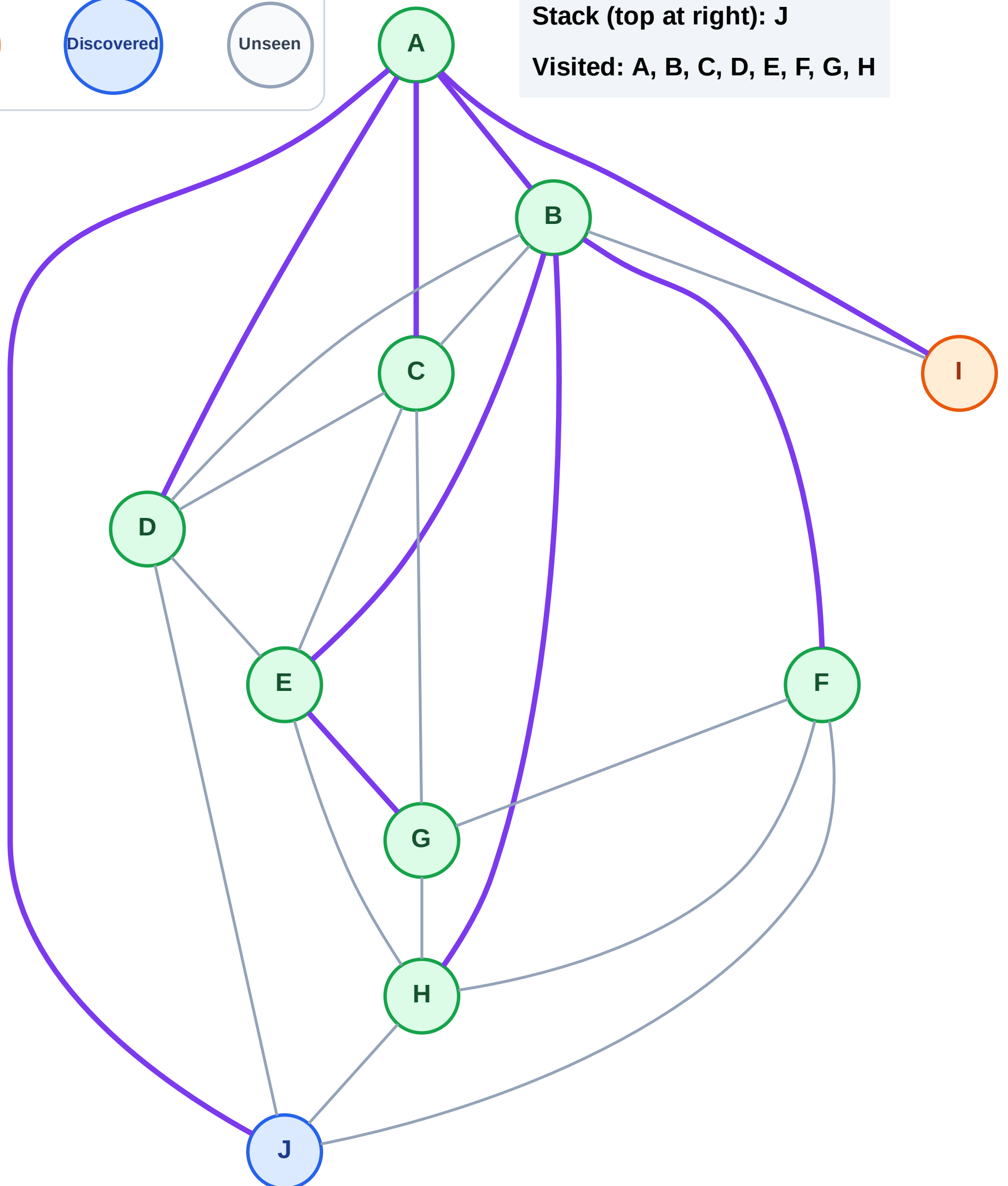
Step 18: Process node I

## Legend



Stack (top at right): J

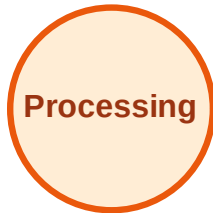
Visited: A, B, C, D, E, F, G, H



# DFS Traversal

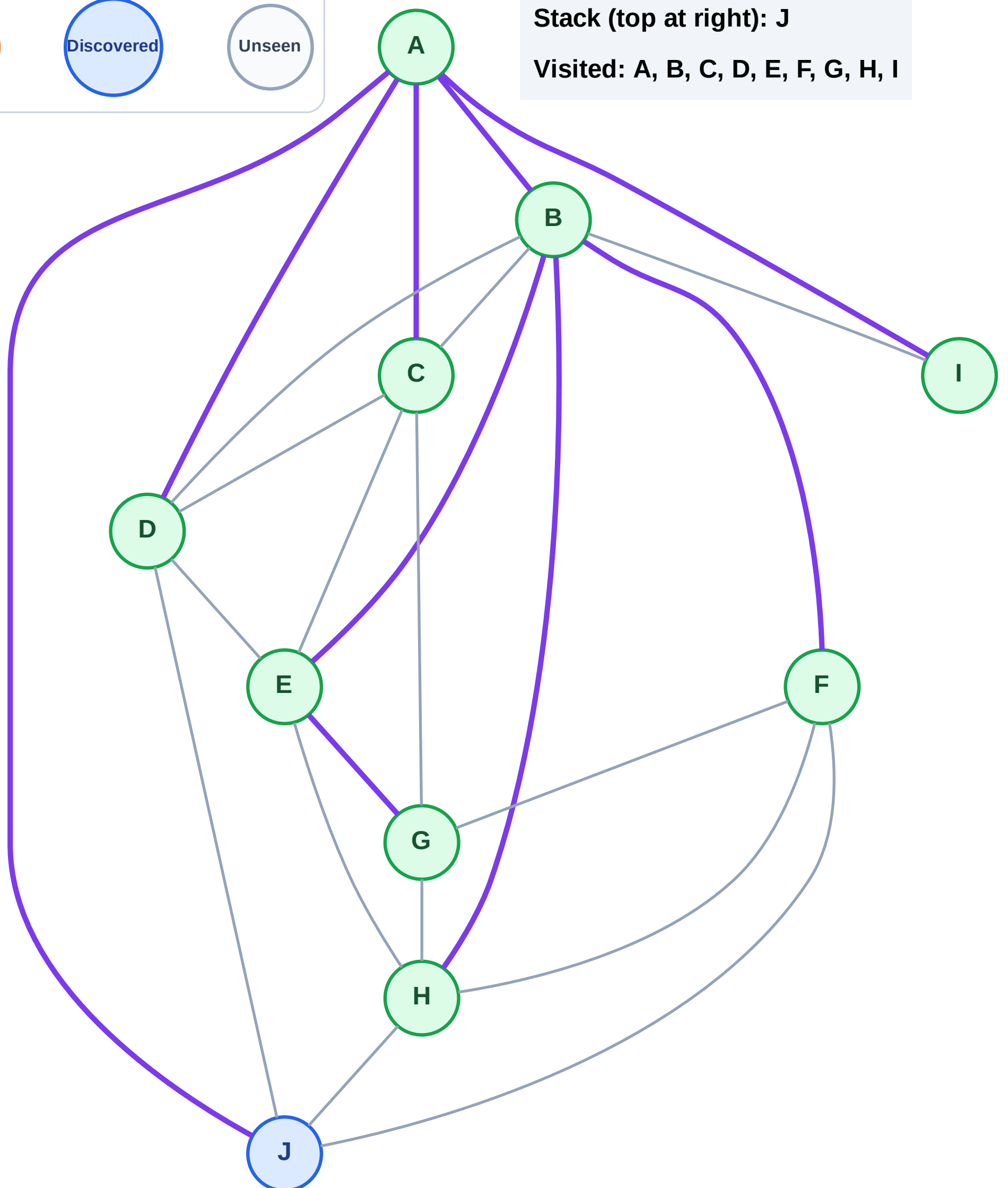
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 20: Process node J

## Legend

Complete

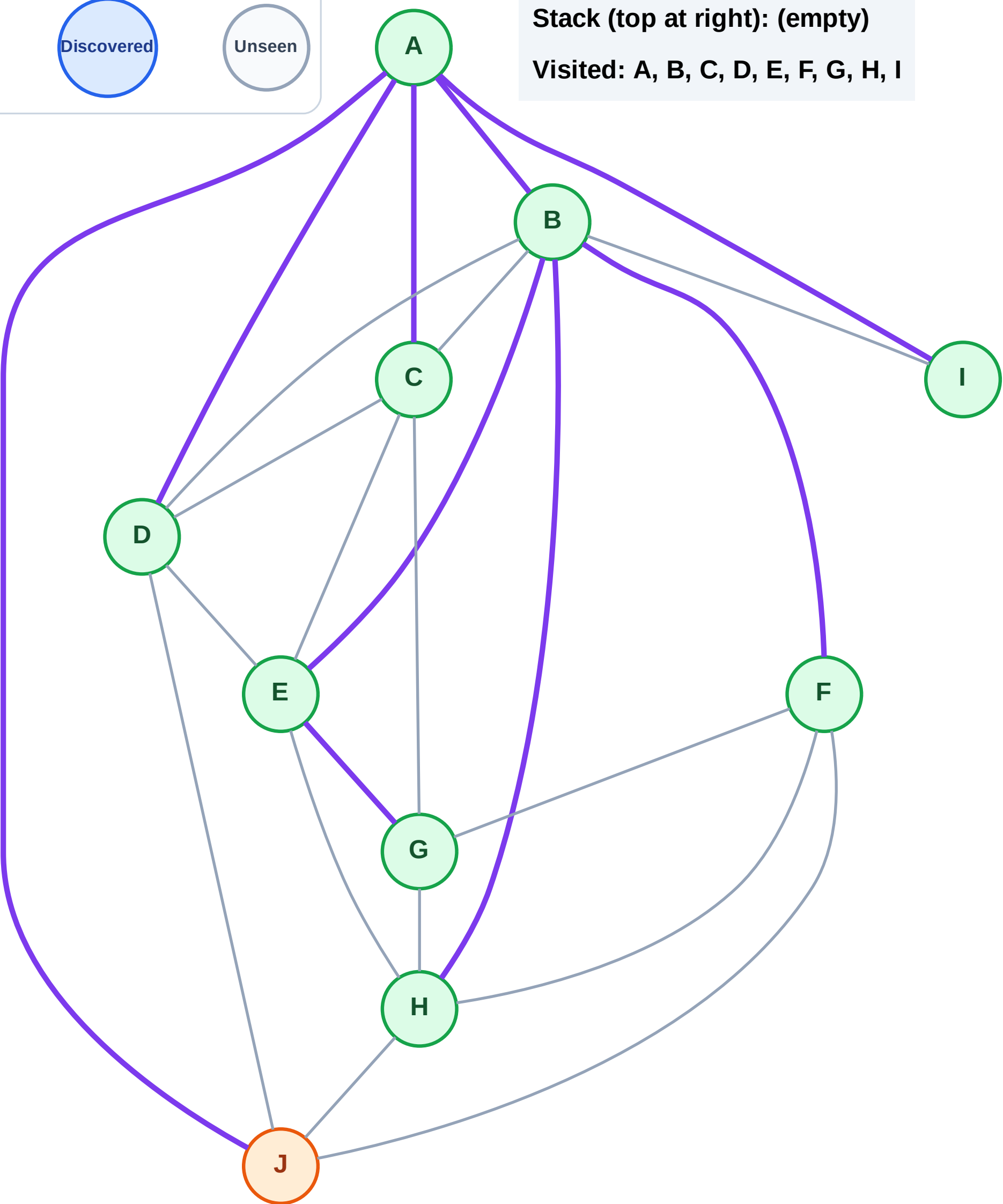
Processing

Discovered

Unseen

Stack (top at right): (empty)

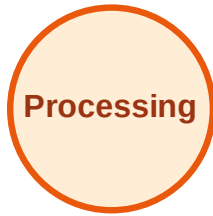
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

